Statistica Sinica

SPARSE WAVELET ESTIMATION OF ADDITIVE FUNCTIONAL COEFFICIENTS

Author(s)

Affiliation(s)

Abstract: We study a varying coefficient model in which the effect of each linear covariate is additive in several modulating covariates, and each additive component is expanded in a periodized wavelet basis. The resulting design is an approximation space of products between a linear covariate and a wavelet evaluated at a modulator, and the components are estimated by a single ~~lasso~~ block lasso problem in which the levels of the coefficients are left unpenalized and the wavelet coefficients of each resolution level are grouped in chunks of order $\log n$. Three results are established. The population Gram matrix of the product design has its smallest eigenvalue bounded away from zero under a conditional second moment condition on the linear covariates and a bounded joint density for the modulators, with no independence between the two groups of covariates; the estimator satisfies a nonasymptotic oracle inequality that needs no cone condition, because the unpenalized levels are profiled out, and that holds for any weights of bounded ratio; and, at a resolution level chosen inside an explicit window, the prediction error and the componentwise error attain the rate of nonlinear wavelet approximation ~~up to a logarithmic factor~~, with no logarithmic factor when the

Besov integrability index is at least two, which is minimax optimal, under the same Besov condition that governs linear approximation. The last result rests on ~~the observation that the Besov condition already implies compressibility of the wavelet coefficients~~a bound on the ideal risk over chunks that the Besov condition already implies, so that adaptivity costs no additional assumption. A threshold on the norms of the fitted components recovers which modulators act on which coefficients, and the thresholded estimator keeps the rate of the fit. The estimator is a single convex program and is fit by ~~coordinate descent on a sparse design matrix~~standard group lasso software. An implementation in R and the code that reproduces every numerical result of the paper are made available with it.

Key words and phrases: Additive coefficient model, Besov space, ~~effective dimension,~~ group lasso, lasso, oracle inequality, thresholding, varying coefficient model, wavelets.

1. Introduction

Applied problems often present an effect that is not constant across the sample. The return to schooling changes with age and with experience, the effect of temperature on electricity demand changes with the hour of the day and with humidity, and the effect of a treatment changes with the age and the body mass index of the patient. The varying coefficient model of Hastie and Tibshirani (1993) and Cai, Fan, and Yao (2000) accommodates this by letting the coefficient of each covariate be a function of a modulating variable; see Fan and Zhang (2008) for a review. When more than one

variable modulates the effect, a fully nonparametric coefficient runs into the curse of dimensionality, and the standard remedy is to make each coefficient additive in the modulators, which is the additive coefficient model of Xue and Yang (2006), studied further by Xue and Liang (2010) and Liu and Yang (2010).

The estimators available for this model are built from polynomial splines and from kernels, and are governed by a global smoothness index, so that a component with a peak, a jump or a change of regime is either over-smoothed or paid for everywhere in the domain. Wavelet bases combined with an ℓ_1 penalty (Tibshirani, 1996) are the standard answer to this difficulty, because a function whose smoothness is inhomogeneous is still economically represented by a few large coefficients (Donoho and Johnstone, 1994, 1998; Antoniadis and Fan, 2001). They have been used for varying coefficients without an additive structure and without penalization (Zhou and You, 2004; Montoril, Morettin, and Chiann, 2018), and for additive models without a multiplying covariate (Sardy and Tseng, 2004; Zhang and Wong, 2003; Haris et al., 2018; Amato et al., 2022; Sardy and Ma, 2024); penalized estimation in the additive coefficient model itself has been carried out with grouped penalties on spline bases (Wei et al., 2011; Antoniadis et al., 2014; Ma et al., 2015), again under global smoothness.

Klopp and Pensky (2015) study the varying coefficient model in which a single index modulates every coefficient. They expand each coefficient function in an orthonormal basis, estimate the resulting array by a block lasso, and obtain a nonasymptotic oracle inequality, a Besov rate that adapts to inhomogeneous smoothness, and a matching minimax lower bound, under the assumption that the covariates are independent of the index. The present paper carries that ~~programme~~ program to the situation an applied problem usually presents, in which several variables modulate a coefficient at once. We let each coefficient be additive in the modulators, which keeps the estimator free of the curse of dimensionality in their number, we allow the covariates to depend on the modulators, and we ~~penalize coefficient by coefficient rather than in blocks, leaving the levels of the coefficients unpenalized. Each of these costs something.~~ keep the block lasso but leave the levels of the coefficients unpenalized and group the wavelet coefficients in chunks of order $\log n$ within a resolution level, with weights whose ratio stays bounded, so that the standardized weights of group lasso software are covered. Additivity produces cross terms between distinct modulators, so the population Gram matrix no longer factorizes as a Kronecker product and has to be bounded directly; dependence between the covariates and the modulators replaces a marginal second moment by a conditional one; and

the unpenalized levels leave a term in the risk that full block penalization does not incur. Our main result, Theorem 1, states that the estimator attains the rate of ~~nonlinear wavelet approximation up to a logarithmic factor, under the same Besov assumption that governs~~ linear approximation, ~~and it reduces to the rate of Klopp and Pensky when a single modulator is present.~~ Klopp and Pensky, with no logarithmic factor when the Besov integrability index is at least two, ~~so that with a single modulator independent of the covariates it is minimax optimal there~~ where it is minimax optimal. It holds under weaker conditions than theirs: the number of covariates is fixed, a regime their high dimensional condition excludes, and the effective smoothness of the components need only exceed $s/(2s+1)$, which is below one half, where theirs must exceed one half. Components with effective smoothness one half, such as the inhomogeneous functions of Section 5, are covered here and not there. ~~A lower bound for several modulators remains open.~~ The lower bound that makes ~~it~~ the rate optimal is proved here for any number of modulators and for covariates that depend on them; Klopp and Pensky have it with a single modulator independent of the covariates.

We call the resulting procedure WAFC, for wavelet additive functional coefficients. Its contributions are of ~~three four~~ three kinds. The first is the theory of the additive product design: a design condition for the Gram

matrix of the products between a linear covariate and a wavelet of a modulator, which carries a cross term between distinct modulators and does not assume independence (Proposition 2); an oracle inequality that needs no cone condition, because the unpenalized levels are profiled out, and that holds for any weights of bounded ratio, including those of the software that computes the estimator (Theorem 2); and rates ~~obtained by compressibility~~ that follow from the Besov assumption alone, through a bound on the ideal risk over chunks (Theorem 1), together with a minimax lower bound, for any number of modulators, that makes these rates optimal when $\pi \geq 2$ (Theorem S6.1 in the Supplementary Material). ~~The second is the reading that compressibility costs no assumption: the Besov condition on the components already implies that their wavelet coefficients lie in a weak ℓ_τ ball with $\tau = (s + 1/2)^{-1}$ (Lemma 3, a form with explicit constants of the classical embedding in Johnstone, 2019, Proposition 9.15), and it is this that separates the rate of the estimator from the rate of linear approximation when the Besov integrability index is smaller than two.~~ The second is the recovery of the structure of the model, which modulators act on which coefficients, by a threshold on the norms of the fitted components (Corollary 2): removing the null blocks needs no separation condition, and the thresholded estimator keeps the rate of the fit (Corollary 3), so that one

estimator serves both for structure and for prediction. The third is computational: the estimator is one convex program ~~on a sparse design matrix~~ solved by standard group lasso software, ~~and~~ the resolution level and the penalty are chosen jointly by cross-validation, and the threshold by the one standard error rule on the same folds. ~~The fourth is the recovery of the structure of the model, which modulators act on which coefficients, by a threshold on the norms of the fitted components (Corollary 2): it needs no design condition beyond the one above and no irrepresentability condition, and removing the null blocks needs no separation condition at all.~~

The rest of the paper is organized as follows. Section 2 introduces the model, its identification, the wavelet approximation space and the estimator. Section 3 states the assumptions and the theory, opening with the main rate and then giving the three results it rests on, then the rates, including one with no design condition, and closes with the recovery of the block structure by thresholding and the risk of the thresholded estimator. Section 4 describes the computation, the scale of the penalty and the tuning rules, the threshold among them. Section 5 compares the method with penalized splines and with other competitors in a simulation study, and Section 6 applies it to hourly air pollution data from London. Proofs are collected in the online Supplementary Material, which also gives the theory

of the coordinatewise lasso, offered by the software as an option, the details and further results of the simulation study, and a second application, to data from Beijing.

Throughout, $\|a\|_\pi$ is the ℓ_π norm of a vector, $\|h\|_n^2 = n^{-1}\sum_{i\le n} h_i^2$ is the empirical norm, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ are the extreme eigenvalues of a symmetric matrix, $\|\cdot\|_{\text{op}}$ is the operator norm, and $a_n \asymp b_n$ means that a_n/b_n is bounded away from zero and infinity. For a support S, δ_S is the vector that agrees with δ on S and vanishes elsewhere.

2. Model and Wavelet Approximation Space

2.1 The additive functional coefficient model

Let Y be a scalar response, $\mathbf{X} = (X_1, \ldots, X_p)^\top$ a vector of linear covariates and $\mathbf{U} = (U_1, \ldots, U_q)^\top$ a vector of modulating covariates, taking values in $[0,1]^q$ after a monotone transformation of each coordinate. The model is

$$Y \;=\; \sum_{\ell=1}^{p} \beta_\ell(\mathbf{U})\, X_\ell \;+\; \varepsilon, \qquad \mathrm{E}(\varepsilon \mid \mathbf{X}, \mathbf{U}) = 0, \tag{2.1}$$

with each functional coefficient additive in the modulators,

$$\beta_\ell(u) \;=\; c_\ell \;+\; \sum_{m=1}^{q} g_{\ell m}(u_m), \qquad \int_0^1 g_{\ell m}(v)\, dv = 0, \qquad \ell = 1, \ldots, p. \tag{2.2}$$

The constant c_ℓ is the level of the ℓth coefficient and $g_{\ell m}$ is the component of that coefficient along the mth modulator. The centering in (2.2) is

with respect to Lebesgue measure, which is the convention of the wavelet literature and the one the basis of Section 2.2 imposes at no cost; the parametrization centered at the law of $\mathbf{U}$ is obtained from it by the shift $c_\ell \mapsto c_\ell + \sum_m \mathrm{E}\{g_{\ell m}(U_m)\}$, which moves mass between the levels and the components but changes neither β_ℓ nor the estimator.

The model contains several familiar ones. With $X_1 \equiv 1$ the first coefficient is an additive intercept, so that $p = 1$ gives the additive model and $p > 1$ with $g_{\ell m} \equiv 0$ for $\ell \geq 2$ gives the partially linear additive model; $q = 1$ gives the classical varying coefficient model with a single index; and $g_{\ell m} \equiv 0$ for all pairs gives linear regression. Writing $f(x, u) = \sum_\ell x_\ell \beta_\ell(u)$ for the regression function, the first question is whether the distribution of $(Y, \mathbf{X}, \mathbf{U})$ determines the parameters. Two degeneracies have to be excluded: if $\mathbf{X}$ is confined to a hyperplane given $\mathbf{U}$, only a combination of the coefficients is determined, and if one modulator is a function of another, only the sum of the corresponding components is.

Proposition 1 (Identification)**.** *Suppose that* $\mathrm{E}\,\|\mathbf{X}\|_2^2 < \infty$*, that* $\mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})$ *is nonsingular almost surely, and that* $\mathbf{U}$ *has a Lebesgue density on* $[0,1]^q$ *that is positive almost everywhere. Then, among the parametrizations that satisfy* (2.2)*, the levels* $c_1, \ldots, c_p$ *and the components* $g_{\ell m} \in L_2[0,1]$ *are determined by* f*: the levels uniquely, and each component up to a Lebesgue*

null set.

Dependence between $\mathbf{X}$ and $\mathbf{U}$ is allowed, and so is dependence among the modulators; what the two conditions exclude is a linear combination of the linear covariates that is a function of the modulators, and a modulator that is a function of the others. Both exclusions return in quantitative form in Section 3.

2.2 The wavelet approximation space

Let ϕ and ψ be the scaling function and the mother wavelet of an orthonormal multiresolution analysis of $L_2(\mathbb{R})$, compactly supported, with ψ bounded and with N vanishing moments, and let L be the length of the filter, so that the support of ~~ψ is contained in $[0, L-1]$~~ϕ is contained in $[0, L-1]$ and that of ψ in $[1-L/2,\ L/2]$, an interval of length $L-1$ (Härdle et al., 1998, Chapter 7). Write

$$\psi_{jk}(u) \;=\; \sum_{\iota\in\mathbb{Z}} 2^{j/2}\psi\{2^j(u+\iota)-k\}, \qquad j\geq 0, \quad k=0,\ldots,2^j-1, \quad (2.3)$$

for the periodization of the basis to $[0,1]$. The system $\{\mathbf{1}\}\cup\{\psi_{jk}\}$ is an orthonormal basis of $L_2[0,1]$ under Lebesgue measure and every wavelet in it integrates to zero (Daubechies, 1992, Section 9.3). We take the coarsest level to be $j_0=0$, so that the only scaling function is the constant $\phi_{00}\equiv 1$,

and we discard it: it is already carried by the level c_ℓ. The approximation space (the sieve) of resolution J is then

$$W_J \;=\; \operatorname{span}\{\psi_{jk} : 0 \le j < J,\; 0 \le k < 2^j\}, \qquad \dim W_J = N_J = 2^J - 1,$$

the orthogonal complement of the constants inside the multiresolution space V_J, and Π_J is the orthogonal projection onto it. Every element of W_J integrates to zero, so that the identification constraint of (2.2) holds by construction, no numerical restriction is imposed on the fit, and the approximation space is a plain linear space.

Lemma 1 (The basis carries the constraint)**.** *Under the conditions of Proposition 1, for every J the map*

$$(\mathbf{c}, \boldsymbol{\theta}) \;\longmapsto\; \sum_{\ell=1}^{p} x_\ell \Big\{ c_\ell + \sum_{m=1}^{q} \sum_{j<J} \sum_{k} \theta_{\ell m, jk} \psi_{jk}(u_m) \Big\}$$

is injective on $\mathbb{R}^p \times \mathbb{R}^d$, $d = pqN_J$; equivalently, the population Gram matrix of the design defined in (2.4) *is positive definite. If instead the column $X_\ell \phi_{00}(U_m)$ is retained in each block, that matrix is singular with nullity exactly pq.*

Replacing $g_{\ell m}$ by $\Pi_J g_{\ell m}$ leaves a bias, which Section 3 controls under a Besov condition. Two practical consequences of periodization deserve mention. A component whose values at the two ends of the interval disagree is

not periodic, and its projection error then stalls at the order $2^{-J/2}$ in L_2 regardless of the interior smoothness (Proposition S3.1 in the Supplementary Material). The two remedies, rescaling the modulators to a proper subinterval and using the basis of Cohen, Daubechies, and Vial (1993) adapted to the interval, are matters of computation and are discussed in Section 4.

2.3 The estimator

Let $(Y_i, \mathbf{X}_i, \mathbf{U}_i)$, $i = 1, \dots, n$, be independent copies of $(Y, \mathbf{X}, \mathbf{U})$. The row of the design matrix is the vector of the p linear covariates followed by the $d = pqN_J$ products between a linear covariate and a wavelet of a modulator,

$$\mathbf{Z} = [\,\mathbf{A}\;\;\mathbf{B}\,], \qquad \mathbf{A}_{i\ell} = X_{i\ell}, \qquad \mathbf{B}_{i,(\ell m,jk)} = X_{i\ell}\,\psi_{jk}(U_{im}), \tag{2.4}$$

with the blocks (ℓ, m) in lexicographic order and, inside a block, the wavelets ordered by level and then by translation. The penalty acts on chunks of wavelet coefficients inside a block. Fix an integer $b \geq 2$ and let j^* be the finest level with $2^{j^*} < b$. In each block (ℓ, m) the levels $j \leq j^*$ form a single chunk, the coarse chunk, and each finer level, which has $2^j \geq b$ coefficients, is cut into $\lfloor 2^j/b \rfloor$ chunks of consecutive translations, all with b coefficients except the last, which also takes the remainder. Write $\mathcal{G}$ for the resulting partition of the d penalized coordinates, $\boldsymbol{\theta}_G$ and $|G|$ for the subvector and the size of a chunk G, and, for positive weights $w = (w_G)_{G \in \mathcal{G}}$,

$\|\boldsymbol{\theta}\|_{\mathcal{G},w} = \sum_{G\in\mathcal{G}} w_G \, \|\boldsymbol{\theta}_G\|_2$. Writing $\mathbf{Y} = (Y_1, \ldots, Y_n)^\top$, the estimator is

[removed in k = 4]

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}}) \;\in\; \operatorname*{arg\,min}_{(\mathbf{c},\boldsymbol{\theta})\in\mathbb{R}^p\times\mathbb{R}^d} \Big\{ \|\mathbf{Y} - \mathbf{A}\mathbf{c} - \mathbf{B}\boldsymbol{\theta}\|_n^2 \;+\; 2\lambda \, \|\boldsymbol{\theta}\|_1 \Big\}.$$

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}}) \;\in\; \operatorname*{arg\,min}_{(\mathbf{c},\boldsymbol{\theta})\in\mathbb{R}^p\times\mathbb{R}^d} \Big\{ \|\mathbf{Y} - \mathbf{A}\mathbf{c} - \mathbf{B}\boldsymbol{\theta}\|_n^2 \;+\; 2\lambda \|\boldsymbol{\theta}\|_{\mathcal{G},w} \Big\}, \tag{2.5}$$

with $b = b_n = \lceil \log n \rceil$ and $w_G = |G|^{1/2}$. Every chunk of a fine level has between b and $2b-1$ coefficients, the coarse chunk has $2^{j^*+1}-1$, which lies between $b-1$ and $2b-3$ (when $J \le j^*+1$ each block is a single chunk), there are at most $pq(1+2^J/b)$ chunks, and the ratio $\varrho = \max_G w_G / \min_G w_G$ of the weights satisfies $\varrho^2 \le (2b-1)/(b-1) \le 3$ (Proposition S5.1 in the Supplementary Material). The theory of Section 3 holds for any weights whose ratio stays bounded, and the bound on ϱ is what makes it cover the weights $|G|^{1/2}$ that group lasso software uses by default. The penalty does not reach the levels $\mathbf{c}$: they are p parameters of a parametric part, and shrinking them would bias every fitted coefficient by a constant. The fitted components and coefficients are recovered from $\widehat{\boldsymbol{\theta}}$ by

$$\widehat{g}_{\ell m}(v) = \sum_{j<J} \sum_{k} \widehat{\theta}_{\ell m, jk} \, \psi_{jk}(v), \qquad \widehat{\beta}_\ell(u) = \widehat{c}_\ell + \sum_{m=1}^{q} \widehat{g}_{\ell m}(u_m).$$

Problem (2.5) is a ~~lasso (Tibshirani, 1996)~~group lasso (Yuan and Lin, 2006) with some coordinates unpenalized, so it is convex, and it is strictly convex, hence uniquely solved, whenever the Gram matrix of $\mathbf{Z}$ is nonsingular. ~~Two structural variants are available at no conceptual cost and are discussed in Section 4: a sparse group lasso (Simon et al., 2013) whose groups are the blocks (ℓ, m), which removes an entire component when a modulator does not act on a coefficient, and the block lasso of Klopp and Pensky (2015).~~ It differs from the block lasso of Klopp and Pensky (2015) in three respects: their chunks run across resolution levels, their weights are equal, and their levels are penalized with everything else. Two other penalties are available and are discussed in Section 4: the lasso (Tibshirani, 1996), with $\|\boldsymbol{\theta}\|_1$ in place of $\|\boldsymbol{\theta}\|_{\mathcal{G},w}$, whose theory is given in Section S8 of the Supplementary Material, and a sparse group lasso (Simon et al., 2013) whose groups are the blocks (ℓ, m). The theory below is for (2.5).

3. Theory

3.1 Assumptions

Assumption 1 (Sampling and error)**.** The observations $(Y_i, \mathbf{X}_i, \mathbf{U}_i)$, $i \leq n$, are independent and identically distributed and follow (2.1), and the error is conditionally sub-Gaussian: there is $\sigma < \infty$ with $\mathrm{E}\{\exp(t\varepsilon_i) \mid \mathbf{X}_i, \mathbf{U}_i\} \leq$

$\exp(\sigma^2 t^2/2)$ almost surely, for all real t.

Assumption 2 (Linear covariates)**.** $\max_\ell |X_\ell| \le B_X$ almost surely, and there are constants $0 < \kappa_1 \le \kappa_2 < \infty$ with $\kappa_1 \le \lambda_{\min}\{\mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \le \lambda_{\max}\{\mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \le \kappa_2$ almost surely.

Assumption 3 (Modulating covariates)**.** $\mathbf{U}$ takes values in $[0,1]^q$ and has a joint Lebesgue density $p_{\mathbf{U}}$ with $0 < c_U \le p_{\mathbf{U}}(u) \le C_U < \infty$ for every $u \in [0,1]^q$.

Assumption 4 (Basis)**.** The wavelets are the periodization (2.3) of an orthonormal, compactly supported family with ψ bounded and N vanishing moments, and $j_0 = 0$.

Assumption 5 (Besov regularity)**.** There are $s > 0$, $1 \le \pi, r \le \infty$ and $C_g < \infty$ such that every component satisfies $\int_0^1 g_{\ell m} = 0$ and, writing $\theta^*_{\ell m, jk} = \langle g_{\ell m}, \psi_{jk} \rangle$ and $\theta^*_{\ell m, j\cdot}$ for the vector of the coefficients of level j,

$$\left\| \theta^*_{\ell m, j\cdot} \right\|_\pi \;\le\; C_g \, 2^{-j(s+1/2-1/\pi)} \qquad \text{for every } j \ge 0.$$

The effective regularity $s' = s - (1/\pi - 1/2)_+$ is positive.

Assumptions 2 and 3 are the quantitative forms of the two conditions of Proposition 1. In the second it is the joint density that is bounded below, not the marginals: if $U_2 = U_1$ almost surely both marginals are uniform

and the design is singular. Assumption 5 is taken as a condition on the sequence of coefficients rather than as membership in a function space, as in Klopp and Pensky (2015); when ψ has more than s vanishing moments and enough Hölder regularity the two are equivalent, on the circle (Cohen, 2003, Section 3.6), and the implication that the proofs use needs only the vanishing moments. The index s' is the regularity that linear approximation sees and s is the one that nonlinear approximation sees; they differ exactly when $\pi < 2$, which is the regime of inhomogeneous smoothness.

3.2 The main result

Write $\widehat{f_i} = \mathbf{A}_i\widehat{\mathbf{c}} + \mathbf{B}_i\widehat{\boldsymbol{\theta}}$ for the fit, ~~d_n for the number of penalized columns at resolution J_n~~ $\mathcal{G}_n$ for the partition of Section 2.3 at resolution J_n with $b = b_n$, and fix a confidence level $\alpha \in (0, 1)$. The penalty is the one that the oracle inequality of Section 3.4 calls for, namely, with $\gamma = \kappa_1 c_U$ and $\Lambda = \kappa_2 C_U + \gamma/2$ the constants of Proposition 2 below,

[removed in k = 4]

$$\lambda_n \;=\; 2\sigma B_X \sqrt{2C_U}\,\sqrt{\frac{2\log(2d_n/\alpha)}{n}} \;\asymp\; \sigma B_X \sqrt{C_U}\,\sqrt{\frac{J_n}{n}}.$$

$$\lambda_n^{\mathcal{G}} \;=\; 2\max_{G\in\mathcal{G}_n}\frac{\bar\lambda_{0,G}}{w_G},\quad \bar\lambda_{0,G} \;=\; \frac{\sigma}{\sqrt{n}}\Big\{B_X\sqrt{2C_U|G|}+\sqrt{2\Lambda\log(|\mathcal{G}_n|/\alpha)}\Big\}. \tag{3.1}$$

For every chunk, $\lambda_n^{\mathcal{G}} w_G \le \varrho\lambda_{n,1}^{\mathcal{G}}$, where $\lambda_{n,1}^{\mathcal{G}} = 2\max_G \bar\lambda_{0,G}$ is of order $\sigma(\log n/n)^{1/2}$, the order of the penalty of the lasso.

[removed in k = 4]

Theorem (Rate of nonlinear approximation)**.** *Let Assumptions 1 to 5 hold with p, q and all the constants fixed as n grows, and set $\tau = (s+1/2)^{-1}$. Suppose $s' > s/(2s+1)$ and take $J_n = \lceil c\log_2 n\rceil$ with*

$$\max\Big\{\frac{\tau}{2},\ \frac{2-\tau}{4s'}\Big\} \;\le\; c \;<\; 1,$$

a window that is nonempty precisely under that condition. Then, with λ_n as in the display above,

$$\big\|\widehat f - f\big\|_n^2 \;=\; O_p\Big\{\Big(\frac{\log n}{n}\Big)^{2s/(2s+1)}\Big\},\qquad \sum_{\ell,m}\|\widehat g_{\ell m}-g_{\ell m}\|_{L_2[0,1]}^2 \;=\; O_p\Big\{\Big(\frac{\log n}{n}\Big)^{2s/(2s+1)}\Big\}.$$

Theorem 1 (Rate of nonlinear approximation)**.** *Let Assumptions 1 to 5 hold with p, q and all the constants fixed as n grows, and let the weights have ratio $\varrho \le \bar\varrho$ for a constant $\bar\varrho$ fixed as n grows ($\bar\varrho^2 = 3$ for the weights of Section 2.3). Suppose $s' > s/(2s+1)$ and take $J_n = \lceil c\log_2 n\rceil$ with*

$$\frac{s}{(2s+1)s'} \;\le\; c \;<\; 1, \tag{3.2}$$

a window that is nonempty precisely under that condition. Then, with $\lambda = \lambda_n^{\mathcal{G}}$ *as in* (3.1) *and* $r_n = n^{-2s/(2s+1)}(\log n)^{(2/\pi-1)_+/(2s+1)}$,

$$\left\|\widehat{f} - f\right\|_n^2 \;=\; O_p(r_n), \qquad \sum_{\ell,m} \left\|\widehat{g}_{\ell m} - g_{\ell m}\right\|_{L_2[0,1]}^2 \;=\; O_p(r_n).$$

In particular, when $\pi \geq 2$ *the rate is* $n^{-2s/(2s+1)}$, *with no logarithmic factor.*

Three readings of Theorem 1 matter. First, the exponent is the one of nonlinear wavelet approximation, $2s/(2s+1)$, and not the one of linear approximation, $2s'/(2s'+1)$ of Theorem 3 below; the two agree when $\pi \geq 2$ and separate when $\pi < 2$, which is where the ℓ_{T} penalty pays for itself. Second, the rate is attained under Assumption 5 alone: ~~no compressibility is assumed on top of it, because the Besov condition already implies it, with $\tau = (s+1/2)^{-1}$, by Lemma 3.~~ no condition is imposed on the coefficients on top of it, because the Besov condition already bounds the ideal risk over chunks that governs the estimation error (Lemma 3). ~~Third, the rate $(\log n/n)^{2s/(2s+1)}$ is the minimax reference of the Gaussian sequence model over Besov bodies with per coordinate noise of order λ_n (Donoho and Johnstone, 1998, Theorems 4 and 5; Johnstone, 2019, Theorem 9.3); the logarithmic factor is the price of the single, nonadaptive penalty level (3.1), which takes a union bound over the d_n columns. No lower bound is proved here for $q \geq 2$: the comparison is with the reference of the sequence~~

~~model, and the minimax lower bound of Klopp and Pensky (2015) covers the case of a single modulator with covariates independent of it.~~ Third, ~~$n^{-2s/(2s+1)}$ is the minimax rate over Besov bodies in the Gaussian sequence model with noise level of order $n^{-1/2}$ (Donoho and Johnstone, 1998, Theorems 4 and 5), and with a single modulator, independent of the covariates, and Gaussian errors, the minimax lower bound of Klopp and Pensky (2015, Theorem 1) holds over the class of Assumption 5 (Remark S6.1 in the Supplementary Material), so that in that case Theorem 1 is minimax optimal when $\pi \geq 2$.~~ the rate cannot be improved. With Gaussian errors and any law of $(\mathbf{X}, \mathbf{U})$ that satisfies Assumptions 2 and 3, the minimax risk over the class of Assumption 5, in the prediction norm and for the components, is at least a constant multiple of $pq\, n^{-2s/(2s+1)}$, for every π (Theorem S6.1 in the Supplementary Material). This is the regression counterpart of the lower bound of Donoho and Johnstone (1998, Theorem 5) for Besov bodies in the Gaussian sequence model, and with a fixed number of covariates it extends that of Klopp and Pensky (2015, Theorem 1), who have a single modulator independent of the covariates. Theorem 1 is therefore minimax optimal when $\pi \geq 2$, with any number of modulators and with covariates that depend on them. When $\pi < 2$ the remaining factor $(\log n)^{(2/\pi-1)/(2s+1)}$ is the one in the rate of Klopp and Pensky (2015, eq. 3.19) with a fixed num-

ber of covariates, and in that of block thresholding in the sequence model (Cai, 1999, Theorem 4); ~~whether it is necessary for this estimator is not known, and no lower bound is proved here for $q \geq 2$.~~ the lower bound carries no such factor, and whether the factor is necessary for this estimator is not known. Compared with the coordinatewise lasso, whose rate under the same conditions is $(\log n/n)^{2s/(2s+1)}$ (Theorem S8.3 in the Supplementary Material), the chunks remove a factor $(\log n)^{2s'/(2s+1)}$. ~~When $s < 1/2$ and $\pi \geq 2$, the rate $n^{-2s/(2s+1)}$ of Theorem 1 is attained with no design condition at all, at another resolution (Proposition 3(ii) below).~~ When $s < 1/2$, the prediction rate of Theorem 1 holds with no design condition at all, under its other assumptions and in the same window for J_n (Proposition S6.2 in the Supplementary Material).

The proof combines three ingredients, given below: the approximation error of W_J, the design condition, and a nonasymptotic oracle inequality. Section 3.5 assembles them, first at ~~a dense reading of the estimation term~~the oracle, which gives the rate of linear approximation, and then at ~~a sparse comparator~~a comparator that keeps only the chunks of large energy, which gives Theorem 1.

3.3 Approximation and the design condition

Lemma 2 (Approximation bias)**.** *Under Assumptions 4 and 5, for every pair* (ℓ, m) *and every* $J \geq 0$,

$$\left\| g_{\ell m} - \Pi_J g_{\ell m} \right\|^2_{L_2[0,1]} \;=\; \sum_{j \geq J} \left\| \theta^*_{\ell m, j\cdot} \right\|_2^2 \;\leq\; \frac{C_g^2}{1 - 2^{-2s'}}\, 2^{-2Js'}.$$

If in addition Assumptions 2 and 3 hold, then, with $f_J(x,u) = \sum_\ell x_\ell \{c_\ell + \sum_m (\Pi_J g_{\ell m})(u_m)\}$,

$$\left\| f - f_J \right\|^2_{L_2(P)} \;\leq\; \frac{(pq)^2 B_X^2 C_U C_g^2}{1 - 2^{-2s'}}\, 2^{-2Js'} \;=:\; \mathcal{B}_J^2. \tag{3.3}$$

The exponent is s' and not s, and this is where the loss $(1/\pi - 1/2)_+$ enters: inside a level, an ℓ_π bound with $\pi < 2$ controls the ℓ_2 norm without gain. The bias uses only the bound on the linear covariates and the marginal density of each modulator, so dependence between $\mathbf{X}$ and $\mathbf{U}$ does not affect it. What it does affect is the passage from the prediction norm to the norm of each component, which is the design condition.

Let $\mathbf{\Psi}(u) \in \mathbb{R}^K$, $K = 1 + qN_J$, collect the constant and the wavelets of the q modulators, so that the row of $\mathbf{Z}$ in (2.4) is a permutation of the Kronecker product $\mathbf{X} \otimes \mathbf{\Psi}(\mathbf{U})$. Write $\Sigma = \mathrm{E}(\mathbf{Z}\mathbf{Z}^\top)$ and $\widehat{\Sigma} = \mathbf{Z}^\top \mathbf{Z}/n$ for the population and empirical Gram matrices and $\Sigma_{\mathbf{\Psi}} = \mathrm{E}\{\mathbf{\Psi}(\mathbf{U})\mathbf{\Psi}(\mathbf{U})^\top\}$ for the part in the modulators alone.

Proposition 2 (Gram matrix of the product design)**.** *Let Assumptions 2 to 4 hold and write* $D = p + d$ *and* $\gamma = \kappa_1 c_U$*.*

(i) $c_U \le \lambda_{\min}(\Sigma_{\mathbf{\Psi}}) \le \lambda_{\max}(\Sigma_{\mathbf{\Psi}}) \le C_U$ *for every* J*, and consequently* $\gamma \le \lambda_{\min}(\Sigma) \le \lambda_{\max}(\Sigma) \le \kappa_2 C_U$*. If* $\mathbf{X}$ *and* $\mathbf{U}$ *are independent,* Σ *is, up to the permutation of columns, the Kronecker product* $\mathrm{E}(\mathbf{X}\mathbf{X}^\top) \otimes \Sigma_{\mathbf{\Psi}}$*, and the two bounds are attained as products of eigenvalues.*

(ii) With $R_J = pB_X^2(1 + qC_\psi 2^J)$*, where* C_ψ *depends only on the wavelet family,* $\|\mathbf{Z}_i\|_2^2 \le R_J$ *almost surely and*

$$\mathrm{P}\Big(\lambda_{\min}(\widehat{\Sigma}) \ge \frac{\gamma}{2}\Big) \;\ge\; 1 - 2D\exp\Big\{-\frac{n\gamma^2}{8R_J(\kappa_2 C_U + \gamma/3)}\Big\},$$

which tends to one whenever $pq2^J \log(pq2^J)/n \to 0$*.*

Part (i) is stated for the general case: independence between the covariates and the modulators buys the exact factorization but not a better constant. A lower bound on the smallest eigenvalue dominates every compatibility and restricted eigenvalue constant, including the versions for groups of Lounici et al. (2011), on every support and with no cone restriction, since $\delta^\top A\delta \ge \lambda_{\min}(A)\,\|\delta\|_2^2 \ge \lambda_{\min}(A)\,\|\delta_S\|_1^2/|S|$, and likewise $\delta^\top A\delta \ge \lambda_{\min}(A)\{\sum_{G\in\mathcal{H}} w_G\,\|\delta_G\|_2\}^2/\mathcal{W}(\mathcal{H})$ for a set $\mathcal{H}$ of chunks, with $\mathcal{W}(\mathcal{H}) = \sum_{G\in\mathcal{H}} w_G^2$, and it is in this form that the next subsection consumes it. In the resolution regime of Section 3.5 the requirement of part

(ii) is ~~of the same order as the rate itself~~the rate times $\log n$, which tends to zero, so it costs nothing there; if pq were larger than $n^{1/2}$, only the compatibility constant could be transferred, through the entrywise deviation (Bühlmann and van de Geer, 2011, Corollary 6.8), at the price of a factor s_0^2 and of the cone.

3.4 A nonasymptotic oracle inequality

Let $\boldsymbol{\beta}^* = (\mathbf{c}, \boldsymbol{\theta}^*)$ be the oracle of the approximation space, that is the vector whose components are the levels and the coefficients $\theta^*_{\ell m,jk}$ of Assumption 5 for $j < J$, so that f_J is its fit, and let

[removed in k = 4]

$$\mathbf{b} = f - f_J \in \mathbb{R}^n, \qquad S_0 = \operatorname{supp}(\boldsymbol{\theta}^*), \qquad s_0 = |S_0|, \qquad v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*.$$

$$\mathbf{b} = f - f_J \in \mathbb{R}^n, \qquad \bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}, \qquad \bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}, \qquad \bar{\mathcal{G}} = \{G : \bar{\boldsymbol{\theta}}_G \neq 0\},$$

for a comparator $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}) \in \mathbb{R}^p \times \mathbb{R}^d$, any vector of levels and coefficients, $\bar{\mathcal{G}}$ being its active chunks, and $\bar{\mathcal{W}} = \mathcal{W}(\bar{\mathcal{G}}) = \sum_{G \in \bar{\mathcal{G}}} w_G^2$. Let $\Pi_{\mathbf{A}}$ be the orthogonal projection onto the column space of $\mathbf{A}$, which exists whenever $\mathbf{A}$ has rank p, let $M_{\mathbf{A}} = I_n - \Pi_{\mathbf{A}}$, and let $\widetilde{\mathbf{B}} = M_{\mathbf{A}}\mathbf{B}$ and $\widetilde{\gamma} = \lambda_{\min}(\widetilde{\mathbf{B}}^\top \widetilde{\mathbf{B}}/n)$ be the residualized design and its smallest eigenvalue, and $\widetilde{\Psi}_G = \widetilde{\mathbf{B}}_G^\top \widetilde{\mathbf{B}}_G/n$

the residualized Gram matrix of the columns of a chunk. Finally let $\widehat{\sigma}^2_{\max} = \max_a(\widehat{\Sigma})_{aa}$ be the largest empirical norm of a penalized column.

[removed in k = 4]

Theorem (Oracle inequality). *Let Assumption 1 hold, suppose* $\mathbf{A}$ *has rank* p*, and take* $\lambda \geq 2\lambda_0$ *with* $\lambda_0 = \sigma\widehat{\sigma}_{\max}\{2\log(2d/\alpha)/n\}^{1/2}$*. On the event* $\mathcal{T} = \{\left\|\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\right\|_\infty \leq \lambda/2\}$*, which has probability at least* $1-\alpha$*:*

(i) with no condition on the design, $\left\|\widetilde{\mathbf{B}}v\right\|_n^2 \leq 6\lambda \left\|\boldsymbol{\theta}^*\right\|_1 + 4\left\|\mathbf{b}\right\|_n^2$*;*

(ii) if $\widetilde{\gamma} > 0$*,*

$$\left\|\widetilde{\mathbf{B}}v\right\|_n^2 \leq \frac{64\lambda^2 s_0}{\widetilde{\gamma}} + 16\left\|\mathbf{b}\right\|_n^2, \qquad \|v\|_2^2 \leq \frac{64\lambda^2 s_0}{\widetilde{\gamma}^2} + \frac{16\left\|\mathbf{b}\right\|_n^2}{\widetilde{\gamma}};$$

(iii) in either regime, $\left\|\widehat{f} - f\right\|_n \leq \left\|\widetilde{\mathbf{B}}v\right\|_n + 2\left\|\mathbf{b}\right\|_n + \left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n$*, and* $\mathrm{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n$*.*

Theorem 2 (Oracle inequality). *Let Assumption 1 hold, suppose* $\mathbf{A}$ *has rank* p*, let the weights not depend on the responses, and, for* $\alpha \in (0,1)$*, let*

$$\lambda_{0,G} = \frac{\sigma}{\sqrt{n}}\left[\{\mathrm{tr}(\widetilde{\Psi}_G)\}^{1/2} + \left\{2\left\|\widetilde{\Psi}_G\right\|_{\mathrm{op}}\log(|\mathcal{G}|/\alpha)\right\}^{1/2}\right]$$

and $\lambda_w = 2\max_{G\in\mathcal{G}}\lambda_{0,G}/w_G$*, and take* $\lambda \geq \lambda_w$*. On the event* $\mathcal{T}_{\mathcal{G},w} = \{\max_G \left\|(\widetilde{\mathbf{B}}^\top\boldsymbol{\varepsilon}/n)_G\right\|_2/w_G \leq \lambda/2\}$*, which has probability at least* $1-\alpha$*, the following hold simultaneously for every comparator* $\bar{\boldsymbol{\beta}}$*:*

(i) with no condition on the design, $\frac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + \lambda\|\widehat{\boldsymbol{\theta}}\|_{\mathcal{G},w} \le 3\lambda\|\bar{\boldsymbol{\theta}}\|_{\mathcal{G},w} + 2\left\|\bar{\mathbf{b}}\right\|_n^2$;

(ii) if $\widetilde{\gamma} > 0$,

$$\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 \le \frac{64\lambda^2\bar{\mathcal{W}}}{\widetilde{\gamma}} + 16\left\|\bar{\mathbf{b}}\right\|_n^2, \qquad \|\bar{v}\|_2^2 \le \frac{64\lambda^2\bar{\mathcal{W}}}{\widetilde{\gamma}^2} + \frac{16\left\|\bar{\mathbf{b}}\right\|_n^2}{\widetilde{\gamma}};$$

(iii) in either regime, $\left\|\widehat{f} - f\right\|_n \le \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n + 2\left\|\bar{\mathbf{b}}\right\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n$, *and* $\mathrm{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) \le \sigma^2 p/n$.

~~Three~~Four features separate Theorem 2 from the standard treatment (Bühlmann and van de Geer, 2011, Chapter 6) and from that of the group lasso by Lounici et al. (2011). The unpenalized levels are not placed inside the support, which would require a compatibility condition on a cone; they are profiled out, and the Schur complement transfers the full smallest eigenvalue of $\widehat{\Sigma}$ to the residualized block, $\widetilde{\gamma} \ge \lambda_{\min}(\widehat{\Sigma})$, so that Proposition 2(ii) is used as it stands and the constant does not depend on ~~S_0~~$\bar{\mathcal{G}}$. The model is misspecified inside the approximation space, and the deterministic residual ~~$\mathbf{b}$~~$\bar{\mathbf{b}}$ is handled by Young's inequality rather than by duality, which keeps the statement nonasymptotic and valid also when the bias dominates, and the statement holds for every comparator at once, which is what Section 3.5 uses. ~~The penalty is calibrated by the empirical column norms and not by $\max_{i,a}|Z_{ia}|$: the former is bounded in probability~~

by $2B_X^2 C_U$, while the latter is of order $2^{J/2}$ and would inflate λ by that factor and destroy the rates of Section 3.5. The penalty is calibrated chunk by chunk, by the tail inequality for quadratic forms of Hsu et al. (2012, Theorem 2.1) with a union bound over the chunks (Lemma S5.2 in the Supplementary Material); the event $\mathcal{T}_{\mathcal{G},w}$ does not depend on the weights, which only choose the chunk that dictates λ_w, and the price of unequal weights is the factor ϱ in $\lambda_w w_G \le 2\varrho \max_{G'} \lambda_{0,G'}$, paid in the estimation term. Since $\operatorname{tr}(\widetilde{\Psi}_G) \le |G| \widehat{\sigma}^2_{\max}$ and $\left\|\widetilde{\Psi}_G\right\|_{\mathrm{op}} \le \lambda_{\max}(\widehat{\Sigma})$, the deterministic level (3.1) dominates λ_w whenever $\widehat{\sigma}^2_{\max} \le 2B_X^2 C_U$ and $\lambda_{\max}(\widehat{\Sigma}) \le \Lambda$, two events of probability tending to one in the regime of Section 3.5; it is the empirical column norms that enter, and not $\max_{i,a} |Z_{ia}|$, which is of order $2^{J/2}$ and would destroy the rates. The software standardizes the columns and orthonormalizes each chunk before penalizing it; the theorem covers that estimator as well, with $\widetilde{\Psi}_G$ replaced by $Q_G^{-1/2} \widetilde{\Psi}_G Q_G^{-1/2}$, Q_G being the centered Gram matrix of the chunk, which makes $\lambda_{0,G}$ pivotal, and with $\bar{\mathcal{W}}$ multiplied by $\max_G \lambda_{\max}(Q_G)$ (Remark S5.2 in the Supplementary Material). The term $\sigma^2 p/n$ in part (iii) is the price of leaving the p levels out of the penalty; no choice of λ removes it, it is of smaller order than the estimation term whenever $p \lesssim s_0 \log(pqN_J) p \lesssim |\bar{\mathcal{G}}| \log n$, and it is absent from Klopp and Pensky (2015), where the levels are penalized along with

everything else.

3.5 Rates

Under Assumption 5 the oracle is generically dense: the coefficients decay but do not vanish, so that ~~$s_0 = d$~~every chunk is active and part (ii) of Theorem 2, read at the oracle, delivers the rate of linear approximation, now with no logarithmic factor.

[removed in k = 4]

Theorem (Rate of linear approximation)**.** *Let Assumptions 1 to 5 hold with p, q and the constants fixed, and take*

$$J_n = \min\{J \in \mathbb{N} : \ 2^J \geq (n/\log n)^{1/(2s'+1)}\},$$

so that $J_n = \log_2 n/(2s'+1) + O(\log\log n)$. Then the requirement of Proposition 2(ii) holds along J_n, being itself of the exact order $(\log n/n)^{2s'/(2s'+1)}$, and, with λ_n as in the first display of Section 3.2,

$$\left\|\widehat{f} - f\right\|_n^2 \ = \ O_p\Big\{\Big(\frac{\log n}{n}\Big)^{2s'/(2s'+1)}\Big\}.$$

At this resolution the estimation term and the bias term are of the same order, and the term $\sigma^2 p/n$ of Theorem 2(iii) is of strictly smaller order.

Theorem 3 (Rate of linear approximation). *Let Assumptions 1 to 5 hold with p, q and the constants fixed, let the weights be as in Theorem 1, and take*

$$J_n = \min\{J \in \mathbb{N} : \ 2^J \geq n^{1/(2s'+1)}\}. \tag{3.4}$$

Then the requirement of Proposition 2(ii) holds along J_n, being of order $n^{-2s'/(2s'+1)} \log n$, and, with $\lambda = \lambda_n^{\mathcal{G}}$ as in (3.1),

$$\left\|\widehat{f} - f\right\|_n^2 \ = \ O_p\big(n^{-2s'/(2s'+1)}\big).$$

At this resolution the estimation term, of order $pq(b_n + 2^{J_n})/n$, and the bias term are of the same order, and the term $\sigma^2 p/n$ of Theorem 2(iii) is of strictly smaller order.

[removed in k = 4]

Corollary (Components and coefficients). *In the setting of Theorem 3, and with r_n denoting the rate of that theorem,*

$$\sum_{\ell,m} \left\|\widehat{g}_{\ell m} - g_{\ell m}\right\|_{L_2[0,1]}^2 = O_p(r_n), \qquad \left\|\widehat{\beta}_\ell - \beta_\ell\right\|_{L_2([0,1]^q)} = O_p(r_n^{1/2}), \quad \ell = 1, \ldots, p,$$

and the same statements hold with the rate of Theorem 1 in place of r_n under the conditions of that theorem.

Corollary 1 (Components and coefficients). *In the setting of Theorem 3, with* $\rho_n^{\mathcal{G}} = n^{-s'/(2s'+1)}$,

$$\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p\{(\rho_n^{\mathcal{G}})^2\}, \qquad \|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} = O_p(\rho_n^{\mathcal{G}})$$

for $\ell = 1, \ldots, p$, *and* $\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 = O_p(\rho_n^{\mathcal{G}})$.

The passage from the prediction norm to the norm of a single component is where the design condition is used a second time, and the smallest eigenvalue appears squared in the constant. In the arguments of this paper the design condition is needed for the components in every case, and for the prediction rate of Theorem 1 only when $s \geq 1/2$. Two further statements complete the picture. The first dispenses with the design condition altogether, at the price of a slower rate when $s \geq 1/2$, and is the unconditional statement of this paper.

[removed in k = 4]

Proposition (Slow rate, no design condition). *Let Assumptions 1, 4 and 5 hold, with* $\mathbf{A}$ *of rank* p *and* λ_n *as in the first display of Section 3.2. Then* $\left\|\widehat{f} - f\right\|_n^2 = O_p\{\lambda_n \|\boldsymbol{\theta}^*\|_1 + \mathcal{B}_{J_n}^2 + p/n\}$, *where* $\|\boldsymbol{\theta}^*\|_1 \leq pqC_g \sum_{j<J_n} 2^{j(1/2-s)}$ *does not depend on* π. *In particular, if* $s > 1/2$ *then* $\left\|\widehat{f} - f\right\|_n^2 = O_p\{(J_n/n)^{1/2}\}$ *for every* $J_n \geq \log_2 n/(4s')$; *and if* $s < 1/2$, *the choice* $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$ *gives* $\left\|\widehat{f} - f\right\|_n^2 = O_p\{(J_n/n)^{2s'/(1-2s+4s')}\}$, *an exponent that equals* $2s/(2s+$

1) *when* $\pi \geq 2$.

Proposition 3 (Slow rate, no design condition)**.** *Let Assumptions 1, 4 and 5 hold, together with the upper bounds in Assumptions 2 and 3 (*$\max_\ell |X_\ell| \leq B_X$, $\lambda_{\max}\{\mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \leq \kappa_2$ *and* $p_{\mathbf{U}} \leq C_U$*), with* $\mathbf{A}$ *of rank* p*, the weights as in Theorem 1,* $\lambda = \lambda_n^{\mathcal{G}}$ *as in* (3.1) *with any constant* $\Lambda > \kappa_2 C_U$*,* $2^{J_n} \leq 2n$ *and* $pq2^{J_n}\log(pq2^{J_n})/n \to 0$*. Then*

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\{\lambda_n^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + \mathcal{B}_{J_n}^2 + p/n\},$$

with $\lambda_n^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \leq \bar{\varrho}\,\lambda_{n,1}^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \leq \bar{\varrho}\,\lambda_{n,1}^{\mathcal{G}}\,\|\boldsymbol{\theta}^*\|_1$*, where* $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = \sum_G \|\boldsymbol{\theta}_G^*\|_2$ *and* $(\lambda_{n,1}^{\mathcal{G}})^2$ *is of order* $\sigma^2 \log n/n$*. Write* $\vartheta_\pi = \min(1 - 1/\pi,\, 1/2)$*.*

(i) If $s > 1/2$ *and* $s' > 1/4$*, then* $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O(1)$ *and* $\left\|\widehat{f} - f\right\|_n^2 = O_p\{(\log n/n)^{1/2}\}$ *for every* $J_n \geq \log_2 n/(4s')$ *that satisfies the conditions above, and one does.*

(ii) If $s < 1/2$ *and* $s' > s/2$*, then* $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O\{1 + b_n^{-\vartheta_\pi} 2^{J_n(1/2-s)}\}$*, and the choice* $2^{J_n} \asymp \{n/b_n^{(2/\pi-1)_+}\}^{1/(1-2s+4s')}$ *gives*

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\{n^{-2s'/(1-2s+4s')}(\log n)^{(2/\pi-1)_+ 2s'/(1-2s+4s')}\},$$

which, when $\pi \geq 2$*, is* $n^{-2s/(2s+1)}$*, the rate of Theorems 1 and 3, with no logarithmic factor.*

The proof uses no lower bound on $\widehat{\Sigma}$*, and no property of* $\widehat{\Sigma}$ *other than* ~~$\lambda_{\max}(\widehat{\Sigma}) \le \Lambda$~~ *an upper bound on* $\lambda_{\max}(\widehat{\Sigma})$*.*

The bound on $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$ is Lemma S6.2 in the Supplementary Material. Through the chain to $\|\boldsymbol{\theta}^*\|_1$, Proposition 3 reproduces the slow rate of the lasso (Proposition S8.1), with the constant $\bar{\varrho}$: a chunk of up to $2b_n - 1$ coefficients enters $\lambda^{\mathcal{G}}_{n,1}$ at the order of the union bound over the chunks. What the chain throws away is the count: at the fine levels $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$ is smaller than $\|\boldsymbol{\theta}^*\|_1$ by the factor $b_n^{-\vartheta_\pi}$, which in (ii) cancels the factor $b_n^{1/2}$ that the size of the chunks puts in $\lambda^{\mathcal{G}}_{n,1}$, entirely when $\pi \ge 2$ and in part when $\pi < 2$; in (i) the coarse chunk dominates both norms and nothing changes. No lower bound on $\widehat{\Sigma}$ enters: the only property of $\widehat{\Sigma}$ that the proof uses is ~~$\lambda_{\max}(\widehat{\Sigma}) \le \Lambda$~~ an upper bound on $\lambda_{\max}(\widehat{\Sigma})$, here $\lambda_{\max}(\widehat{\Sigma}) \le \Lambda$, which makes the deterministic level (3.1) admissible and holds with probability tending to one under the upper bounds of Assumptions 2 and 3 alone, so that neither c_U nor κ_1 is needed. The conditions $s' > 1/4$ in (i) and $s' > s/2$ in (ii) bite only when $\pi < 2$, and make the resolution of the balance admissible. The pairs that these conditions leave out, $s' \le 1/4$ with $s > 1/2$ and $s' \le s/2$ with $s < 1/2$, all with $\pi < 2$, and the boundary $s = 1/2$ call for a resolution above $n/\log n$, where the column norms and $\lambda_{\max}(\widehat{\Sigma})$ grow with $2^{J_n}/n$; Proposition S6.1 in the Supplementary Material covers them with

the level inflated to $\lambda_n^{\mathcal{G},+} = (\mu_{J_n}^{\mathcal{G}})^{1/2}\lambda_n^{\mathcal{G}}$, where $\mu_J^{\mathcal{G}} = 1 + c_\mu 2^J \log(2D)/n$ for a constant c_μ given there, so that the first display of Proposition 3, with $\lambda_n^{\mathcal{G},+}$ in place of $\lambda_n^{\mathcal{G}}$, holds for every $s' > 0$. There, too, the only property of $\widehat{\Sigma}$ that the proof uses is an upper bound, $\lambda_{\max}(\widehat{\Sigma}) \le \Lambda\mu_{J_n}^{\mathcal{G}}$, which holds with probability tending to one with no condition on J_n and, again, without c_U or κ_1. When $s < 1/2$ the slow rate improves further. Read at the comparator that keeps the chunks with $\|\boldsymbol{\theta}_G^*\|_2 > \lambda w_G$ and sets the others to zero, Theorem 2(i) charges the penalty only for the chunks kept and leaves the energy of the others to the bias, where an upper bound on $\lambda_{\max}(\Sigma)$ controls it; with $s' > s/(2s+1)$ and J_n in the window (3.2), this gives the prediction rate of Theorem 1 with no design condition (Proposition S6.2 in the Supplementary Material). When $s \ge 1/2$ no comparator does better than the order $\lambda_{n,1}^{\mathcal{G}} \asymp (\log n/n)^{1/2}$ of part (i): a coarse coefficient of fixed size costs its product with $\lambda_{n,1}^{\mathcal{G}}$ in the penalty or its square in the bias. Neither reading reaches the components, which need the design condition (Corollary 1).

The second is the approximation lemma that produces Theorem 1. ~~For a vector $\boldsymbol{\theta}$ let $\theta_{(1)} \ge \theta_{(2)} \ge \cdots$ be the nonincreasing rearrangement of the absolute values of its entries and let $\boldsymbol{\theta}^{(M)}$ retain the M largest of them.~~ For $\boldsymbol{\theta} \in \mathbb{R}^d$ and $\eta > 0$, let $R_{\mathcal{G}}(\boldsymbol{\theta}; \eta) = \sum_{G\in\mathcal{G}} \min(\|\boldsymbol{\theta}_G\|_2^2, \eta)$ be the ideal

risk over chunks: the risk of an oracle that keeps a chunk when its energy exceeds η, at the price η, and drops it otherwise.

[removed in k = 4]

Lemma (Besov regularity implies compressibility)**.** *Under Assumptions 4 and 5 with $\pi(s+1/2) > 1$, the oracle satisfies, for every J and every $k \geq 1$,*

$$\theta^*_{(k)} \;\leq\; C_\tau\, k^{-1/\tau}, \qquad \tau = \frac{1}{s+1/2}, \qquad C_\tau = C_g (pq\, A_{s,\pi})^{s+1/2}, \qquad A_{s,\pi} = 1 + \frac{1}{1 - 2^{1-\pi(s+1/2)}},$$

with constants that do not depend on J and with $\tau \in (0,2)$ for every $s > 0$. Consequently $\sum_{k>M} \theta^{*2}_{(k)} \leq \tau(2-\tau)^{-1} C_\tau^2 M^{1-2/\tau}$ for every $M \geq 1$.

Lemma 3 (Besov regularity bounds the ideal risk over chunks)**.** *Under Assumptions 4 and 5, let $b \geq 1$ and let $\mathcal{G}$ be a partition in which, in each block, the levels with $2^j < b$ form at most one chunk and every level with $2^j \geq b$ is cut into chunks contained in it with at least b coefficients each, as the partition of Section 2.3 is. Then, for every J and every $\eta > 0$,*

$$R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) \;\leq\; pq \Big\{ A^{\mathcal{G}}_{s,\pi}\, C_g^\tau \Big(\frac{\eta}{b}\Big)^{2s/(2s+1)} b^{(2/\pi - 1)_+/(2s+1)} + \eta \Big\}, \qquad \tau = \frac{1}{s+1/2},$$

with $A^{\mathcal{G}}_{s,\pi} = 2 + \{1 - 2^{1-\pi(s+1/2)}\}^{-1}$ if $\pi \leq 2$ and $A^{\mathcal{G}}_{s,\pi} = 2 + (1 - 2^{-2s})^{-1}$ if $\pi \geq 2$.

~~The inclusion of Besov balls in weak ℓ_τ balls is classical (Johnstone, 2019, Proposition 9.15); what the lemma adds is the constant C_τ, uniform in J, for the oracle of~~

the product design, which is the form in which the proof of Theorem 1 uses it. The exponent is the one of Lemma 4 in the supplementary material of Klopp and Pensky (2015), but the assumption is weaker: their class is a weighted ℓ_ν ball over all the coefficients, which implies Assumption 5 level by level, and Assumption 5 does not need the sum over the levels. The coarse chunk costs at most one η per block.

Theorem 1 follows by reading Theorem 2(ii) not at the oracle but at the best M term approximation of it, which is legitimate because the proof of that theorem uses the comparator only through the decomposition of $\mathbf{Y}$ and the size of its support. Balancing the two terms that result, $\lambda^2 M/\widetilde{\gamma}$ from estimation and $M^{1-2/\tau}$ from the truncation, gives the effective dimension

[removed in k = 4]

$$M_n^\star \asymp \left(\frac{n}{J_n}\right)^{\tau/2} \ll d_n \asymp pq\, 2^{J_n},$$

and a bound of order $\lambda_n^{2-\tau}$, which is the rate of Theorem 1. The three inequalities that define the window (3.2) are, in order, that the minimum in M be interior, that the approximation bias not dominate, and that the requirement of Proposition 2(ii) hold. Nothing in this argument says that the estimator selects $M_n^\star$ coordinates: no minimal signal or irrepresentability condition is used, and what compressibility buys is a rate governed by $M_n^\star$

~~rather than by d_n, not the recovery of a support.~~ Theorem 1 follows by reading Theorem 2(iii) not at the oracle but at the comparator that keeps the chunks of $\boldsymbol{\theta}^*$ whose squared norm exceeds a level η and sets the others to zero, which the theorem allows because it holds for every comparator at once. The estimation term then counts η for each chunk kept, and the truncation leaves the energy of the others, so that the bound is of the order of $R_{\mathcal{G}}(\boldsymbol{\theta}^*;\eta)$ (Corollary S5.1 in the Supplementary Material), with η of the order of $\max_G(\lambda w_G)^2$, that is of b_n/n. Lemma 3 with $b = b_n$ turns this into $n^{-2s/(2s+1)} b_n^{(2/\pi-1)_+/(2s+1)}$: the chunk pays b_n in the level η and returns it through $(\eta/b)^{2s/(2s+1)}$, so that the cost is of order $1/n$ per coefficient, against $\log n/n$ for the lasso, and the logarithm of the union bound over the columns is absorbed by the size of the chunks, which is why they must have at least of the order of $\log n$ coefficients. When $\pi < 2$ the bound charges $b_n^{(2/\pi-1)/(2s+1)}$ to sequences whose large coefficients sit one per chunk; a signal spread over a level pays nothing for the chunks. The two inequalities that define the window (3.2) are that the approximation bias not dominate and that the requirement of Proposition 2(ii) hold. Nothing in this argument says that the estimator selects the chunks kept by the comparator: no minimal signal or irrepresentability condition is used. The recovery that matters for the model is not of coordinates but of blocks, and it is taken up next.

3.6 Which modulators act on which coefficients

The structure of the model is the set of blocks whose component does not vanish,

$$\mathcal{S} \;=\; \big\{(\ell, m) :\; \nu_{\ell m} > 0\big\}, \qquad \nu_{\ell m} = \big\|g_{\ell m}\big\|_{L_2[0,1]},$$

and to know it is to know which modulators act on which coefficients. The ~~lasso~~ estimator (2.5) is not claimed to recover it: its penalty removes chunks and not blocks, under Assumption 5 the oracle is dense, and nothing prevents the fit from keeping nonzero coefficients in blocks whose component is zero. What the preceding results do give is that every block is estimated at the rate of Corollary 1, and that is enough to recover $\mathcal{S}$ in a second step, by a threshold on the norms of the fitted components. The procedure is estimation followed by a threshold. By the orthonormality of the basis, the norm of a fitted component is the Euclidean norm of its block of coefficients,

$$\widehat{\nu}_{\ell m} = \big\|\widehat{g}_{\ell m}\big\|_{L_2[0,1]} = \Big\|\widehat{\boldsymbol{\theta}}_{\ell m,\cdot}\Big\|_2, \qquad \widehat{\mathcal{S}}(t) = \big\{(\ell, m) :\; \widehat{\nu}_{\ell m} > t\big\},$$

so that the statistic costs nothing once the estimator has been fit. Write ~~$\rho_n = (\log n/n)^{s'/(2s'+1)}$~~ $\rho_n^{\mathcal{G}} = n^{-s'/(2s'+1)}$ for the square root of the rate of Theorem 3.

Assumption 6 (Block separation)**.** The components may depend on n, within the class of Assumption 5 and with its constants fixed, and so may

$\mathcal{S}$; the smallest active block, $\nu_{\min,n} = \min_{(\ell,m)\in\mathcal{S}} \nu_{\ell m}$, with $\nu_{\min,n} = \infty$ when $\mathcal{S}$ is empty, satisfies ~~$\nu_{\min,n}/\rho_n \to \infty$~~ $\nu_{\min,n}/\rho_n^{\mathcal{G}} \to \infty$.

Corollary 2 (Block structure by thresholding)**.** *Let the assumptions of Theorem 3 hold, with J_n and ~~λ_n~~ $\lambda = \lambda_n^{\mathcal{G}}$ as there.*

(i) (Screening.) If the thresholds satisfy ~~$t_n/\rho_n \to \infty$~~ $t_n/\rho_n^{\mathcal{G}} \to \infty$, then $\mathrm{P}\{\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}\} \to 1$.

(ii) (Recovery.) If, in addition, Assumption 6 holds and $\limsup_n t_n/\nu_{\min,n} < 1$*, then* $\mathrm{P}\{\widehat{\mathcal{S}}(t_n) = \mathcal{S}\} \to 1$*. Such thresholds always exist; one is ~~$t_n = (\rho_n \nu_{\min,n})^{1/2}$~~ $t_n = (\rho_n^{\mathcal{G}} \nu_{\min,n})^{1/2}$.*

(iii) (A window at fixed n.) With α the level in (3.1), for every n and every $\alpha' \in (0,1)$ there is an event, of probability at least $1 - \alpha - \alpha' - \omega_n$ with $\omega_n \to 0$ not depending on the components, on which $\widehat{\mathcal{S}}(t) = \mathcal{S}$ for every t with ~~$\bar{\Delta}_n \le t < \nu_{\min,n} - \bar{\Delta}_n$~~ $\bar{\Delta}_n^{\mathcal{G}} \le t < \nu_{\min,n} - \bar{\Delta}_n^{\mathcal{G}}$; here ~~$\bar{\Delta}_n \le c(\alpha,\alpha')\rho_n$~~ $\bar{\Delta}_n^{\mathcal{G}} \le c(\alpha,\alpha')\rho_n^{\mathcal{G}}$ is an explicit bound on $\max_{(\ell,m)} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, given in the proof (Section S7 of the Supplementary Material).

Parts (i) and (ii) hold also at the resolution of Theorem 1, with ~~ρ_n~~ $\rho_n^{\mathcal{G}}$ replaced by ~~$\tilde{\rho}_n = (\log n/n)^{s/(2s+1)}$~~ $\tilde{\rho}_n^{\mathcal{G}} = n^{-s/(2s+1)} (\log n)^{(2/\pi-1)_+/\{2(2s+1)\}}$, the square root of the rate of that theorem, both in the condition on t_n

and in Assumption 6. Since $\tilde{\rho}_n \leq \rho_n$ $\tilde{\rho}_n^{\mathcal{G}} \leq \rho_n^{\mathcal{G}}$ for large n, with a gap in the exponent when $\pi < 2$, the separation required is weaker exactly where the components are spatially inhomogeneous. Part (iii) is stated at the resolution of Theorem 3, where its bound is explicit. The componentwise rate of the lasso at its own resolution is $(\log n/n)^{s'/(2s'+1)}$ (Corollary S8.1 in the Supplementary Material), so that the separation required here is weaker than for the lasso by the factor $(\log n)^{s'/(2s'+1)}$.

Part (i) uses no separation: removing the null blocks is free, and Assumption 6 is spent entirely on not losing a small active block. Under a fixed target $\nu_{\min,n}$ is a positive constant and the assumption holds automatically, because $\rho_n \to 0$ $\rho_n^{\mathcal{G}} \to 0$; it carries content in a triangular array, where it says how fast the smallest active block may shrink. It bears on the components and not on $(\mathbf{X}, \mathbf{U})$, so that no design condition beyond Proposition 2 and no irrepresentability condition enters; a separation condition on the components plays the same role in the selection of variables in nonparametric additive models (Huang et al., 2010). The proof rests on a deterministic sandwich (Lemma S7.1 in the Supplementary Material): if every component is estimated within $\bar{\Delta}$ in $L_2[0,1]$, then $\{\nu_{\ell m} > t + \bar{\Delta}\} \subseteq \widehat{\mathcal{S}}(t) \subseteq \{\nu_{\ell m} > t - \bar{\Delta}\}$, so that every $t \in [\bar{\Delta}, \nu_{\min,n} - \bar{\Delta})$ recovers $\mathcal{S}$, and that window cannot be widened by the argument. Two

things are left out. The window involves σ, the design constant and s', so the corollary does not say how to choose t_n from the data (Section 4.4 uses the one standard error rule of cross-validation, for which no theorem is given here); and it says nothing about which wavelets are active inside a block.

Thresholding is also the estimator used for prediction (Section 4.4). Write $\widehat{\boldsymbol{\theta}}^{(t)}$ for $\widehat{\boldsymbol{\theta}}$ with the blocks outside $\widehat{\mathcal{S}}(t)$ set to zero, and $\widehat{g}^{(t)}_{\ell m}$ and $\widehat{f}^{(t)} = \mathbf{A}\widehat{\mathbf{c}} + \mathbf{B}\widehat{\boldsymbol{\theta}}^{(t)}$ for the resulting components and fit, so that the levels and the blocks kept are those of the fit. Setting to zero an active block whose fitted norm is at most t costs at most its norm, which is at most $t + \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, setting to zero a null block only reduces the error, and the blocks kept do not change (Lemma S7.2 in the Supplementary Material). With the rates of the fit this gives the risk of the thresholded estimator.

Corollary 3 (Risk of the thresholded estimator)**.** *Let $(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}})$ be either the fit of Theorem 3, with $r_n = \rho_n^{\mathcal{G}}$, or the lasso at the resolution of Theorem S8.2 in the Supplementary Material, with $r_n = (\log n/n)^{s'/(2s'+1)}$, and let $t_n \geq 0$.*

(i) With no further assumption, $\sum_{\ell,m} \|\widehat{g}^{(t_n)}_{\ell m} - g_{\ell m}\|^2_{L_2[0,1]} = O_p(r_n^2 + t_n^2)$ and $\left\|\widehat{f}^{(t_n)} - f\right\|_n^2 = O_p(r_n^2 + t_n^2)$.

(ii) If Assumption 6 holds with r_n in place of $\rho_n^{\mathcal{G}}$ and $\limsup_n t_n/\nu_{\min,n} < 1$, then with probability tending to one $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t_n)$,

$$\sum_{\ell,m} \|\widehat{g}_{\ell m}^{(t_n)} - g_{\ell m}\|_{L_2[0,1]}^2 \;\le\; \sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2$$

and

$$\left\| \widehat{f}^{(t_n)} - f \right\|_n^2 \;\le\; \frac{4\Lambda}{\gamma} \left\| \widehat{f} - f_{J_n} \right\|_n^2 + 2\,\|\mathbf{b}\|_n^2 ,$$

so that the thresholded estimator keeps the rate r_n^2 of the fit. If moreover $t_n/r_n \to \infty$, then $\widehat{\mathcal{S}}(t_n) = \mathcal{S}$ with probability tending to one, and the null blocks no longer contribute to the error of the components.

Under a fixed target Assumption 6 holds automatically, and (ii) holds for every $t_n \to 0$. The factor $4\Lambda/\gamma$ in (ii) bounds the condition number of $\widehat{\Sigma}$, through which the bound passes from prediction to coefficients and back; it is a price of the argument, and the corollary says what any threshold costs, not which threshold a data driven rule selects.

4. Computation and Tuning

4.1 The design matrix and the solver

The design (2.4) is built one block at a time: the $n \times N_J$ matrix of the wavelets of U_m is evaluated once per modulator, by the cascade algorithm of Daubechies and Lagarias (1991) as implemented in the `WaveBased` package

(Montoril, 2026), and then multiplied rowwise by X_ℓ. Since a compactly supported wavelet of level j is nonzero on an interval of length $(L-1)2^{-j}$, at most $L-1$ of the 2^j columns of a level are nonzero at a given observation, so a block has at most $n(L-1)J$ nonzero entries out of nN_J and the design is stored in sparse form as soon as J is moderate. Evaluating the basis once on a grid and looking the values up removes the cascade from the inner loop of a simulation.

Problem (2.5) is then a ~~lasso~~group lasso with p unpenalized coordinates, which ~~coordinate descent~~group descent (Breheny and Huang, 2015) solves along a path of penalty levels; we use the ~~glmnet~~`grpreg` package ~~(Friedman et al., 2010)~~ (Breheny and Miller, 2026) of R (R Core Team, 2026). Two points of scale deserve care, because they are what ties the software to the theory. First, the solver standardizes the columns and orthonormalizes each chunk before penalizing it (Simon and Tibshirani, 2012), so that what it computes is (2.5) with $\|\boldsymbol{\theta}_G\|_2$ replaced by $\left\|Q_G^{1/2}\boldsymbol{\theta}_G\right\|_2$, where Q_G is the Gram matrix of the columns of the chunk after centering, which is the estimator of Remark S5.2 in the Supplementary Material; and its objective, $(2n)^{-1}\|\mathbf{Y}-\mathbf{A}\mathbf{c}-\mathbf{B}\boldsymbol{\theta}\|_2^2+\lambda\sum_G w_G\left\|Q_G^{1/2}\boldsymbol{\theta}_G\right\|_2$, is half that of (2.5), so that the penalty level it reports is the λ of (2.5), with no conversion. Second, the solver always fits an intercept of its own, which carries

the level of the constant covariate when there is one. For the lasso option, solved by coordinate descent with the `glmnet` package (Friedman et al., 2010), the penalty level reported for a fit is the λ of ~~(2.5)~~the lasso problem and not the one of the solver, which rescales its own by the ratio between the total number of columns and the number of penalized ones when some coordinates carry zero penalty~~; the conversion is checked by verifying the Karush-Kuhn-Tucker conditions of (2.5) at every point of the path. Second,~~, and when one linear covariate is the constant, its column has zero variance, and the solver may either drop it or share the level with its own intercept; the fit is the same either way, but the level c_ℓ of the constant covariate is only recovered after the intercept is added back to it. For both penalties the scale is checked by verifying the Karush-Kuhn-Tucker conditions of the problem at every point of the path.

4.2 Rescaling and the boundary

The theory places $\mathbf{U}$ in $[0,1]^q$ by assumption. In practice each modulator is mapped to that interval by a monotone transformation~~, and,~~. By default the image is $[0,1]$ itself, which is the setting of Section 3. Optionally, as in the wavelet estimators of Montoril, Chang, and Vidakovic (2019), the image is $[\epsilon, 1-\epsilon]$ with ϵ ~~of the order 2^{-J}~~fixed, which keeps the observations away

from the ~~strip~~point where the periodization of (2.3) ~~leaves an artifact~~joins the two ends of the interval; the price is that the fitted component integrates to zero over the rescaled range and not over $[0, 1]$, a shift of level and not of shape. The margin is a finite sample device, and the reason to allow it is approximation: with the data away from the ends, a component needs to be regular only on its support, where it can be replaced by a periodic extension that matches at $0 \equiv 1$, built from a bounded extension operator (Triebel, 1983, Chapter 3). This removes the stall of Proposition S3.1 and restores the bias of Lemma 2, with a Besov constant of order $\epsilon^{-(s-1/\pi)}$, an order that, by the embedding of Besov into Hölder spaces, cannot be improved when the component takes different values at the two ends of its support. Two features keep the margin out of the theory. It has to be fixed, since with ϵ of the order 2^{-J} the exponent guaranteed for the bias is no better than that of periodization without a margin; and it degrades the design, since the wavelets supported inside the margin are invisible to the data, and the smallest eigenvalue of the Gram matrix of the basis restricted to the support of the data vanishes as soon as $2^J \geq (L-1)/(2\epsilon)$, so that Proposition 2 does not hold as stated. The alternative is the basis adapted to the interval of Cohen, Daubechies, and Vial (1993), which reproduces polynomials on $[0, 1]$ and characterizes Besov regularity without any periodicity (Cohen,

2003, Section 3.9), so that Lemma 2 holds with the Besov space of the interval and the same proof. Its cost is structural: the coarsest level must be large enough for the two boundary zones to be disjoint, the constant is no longer a single column but a combination of the scaling functions, and the constraint of (2.2) has to be imposed by reparametrization, which leaves $pq(2^{j_0} - 1)$ unpenalized parameters instead of none. We therefore keep the periodized basis as the default and offer the interval basis as an option.

4.3 Choosing the resolution and the penalty

The pair (J, λ) is chosen jointly. The default is K-fold ~~cross validation~~cross-validation on a grid of resolutions, with the folds drawn once and reused for every J, one design and one penalty path per resolution, and the usual two readings of the curve: the minimizer of the ~~cross validated~~cross-validated error and the largest penalty within one standard error of it; the threshold of Section 4.4 is applied to the fit at the minimizer. In the software the whole procedure, from the cross-validation over (J, λ) to the threshold, is the single call `cv.wafc(x, u, y)`. Two cheaper rules are also available for the lasso option, the Bayesian information criterion and its extended version, with degrees of freedom equal to the number of nonzero coefficients plus the p levels, which is the usual unbiased count for the lasso.

The theory supplies a fourth rule, since (3.4) and (3.1) are explicit. It is not a practical competitor as it stands, because it needs σ, which has to be estimated, and s', which is unknown, and because it is a statement about order~~, so that at moderate sample sizes it can sit above the resolution that minimizes the realized error, the pre-asymptotic regime being wide when s' is small~~. In our experiments with the lasso option the penalty ~~(3.1)~~ of its theory is 7 to 13 times the one that minimizes the realized error, and it is the penalty, more than the resolution, that makes the rule costly; the resolution ~~(3.4)~~ of its theory misses the best one by up to two levels, above or below according to s'. Its interest is as a reference for what the ~~cross validated~~cross-validated choice ought to approach~~, and the difference between the two is a finite sample quantity worth measuring~~. ~~The four rules are compared in the simulation study.~~

A remark on the ~~structural variant~~other penalties. The lasso option, with $\|\boldsymbol{\theta}\|_1$ in place of $\|\boldsymbol{\theta}\|_{\mathcal{G},w}$, is fit by `glmnet`; its theory is in Section S8 of the Supplementary Material, where its rates carry an extra logarithmic factor, and the threshold of Section 4.4 applies to it unchanged. Replacing ~~$\|\boldsymbol{\theta}\|_1$~~$\|\boldsymbol{\theta}\|_{\mathcal{G},w}$ in (2.5) by the sparse group lasso penalty of Simon et al. (2013), with one group per block (ℓ, m) and fit with the `sparsegl` package (Liang et al., 2024), adds a second tuning parameter and removes entire

components rather than isolated ~~coefficients~~chunks, which is what a reader interested in knowing which modulators act on which coefficients would want, although for ~~the lasso itself~~ (2.5) that question is already answered by thresholding the block norms (Corollary 2). It is a different estimator, with a different oracle inequality (Lounici et al., 2011), and the theory above does not cover it; the choice between the two is settled empirically.

4.4 Choosing the threshold

The threshold is chosen on the folds of Section 4.3, at the pair (J, λ) already selected: each fold is refitted at that pair, with no new search, thresholded at every candidate t and scored on the observations it left out. A thresholded fit is constant in t between two consecutive block norms, so the candidates are exhaustive and not a grid. The default is the largest t whose cross-validated error is within one standard error, between folds, of the smallest, the one standard error rule of the penalty carried to the threshold; the minimizer of the cross-validated error is available as an option aimed at prediction. The fit is not refitted after the threshold: the levels and the blocks kept are those of the fit, which is the estimator of Corollaries 2 and 3. Neither rule has a theorem here: Corollary 2 gives the window of thresholds that recovers the structure, and Corollary 3 what any threshold

costs.

5. Simulation

5.1 Design

The samples are drawn from model (2.1) with $X_1 \equiv 1$, the other linear covariates independent and standard normal, the modulators independent of $\mathbf{X}$, independent of each other and uniform on $[0, 1]$, and Gaussian errors. The levels are $(c_1, c_2, c_3, c_4) = (1, 2, -1.5, 0.5)$, repeated cyclically when $p > 4$, and each nonzero component is a test function centered and scaled to have unit norm in $L_2[0, 1]$. Unless it is fixed, the standard deviation of the error is that of the regression function over the sample divided by a signal to noise ratio.

The core design has five cells. In the smooth cell, with $(p, q) = (3, 2)$, the first coefficient is additive in the two modulators, with g_{11} and g_{12} proportional to $\sin(2\pi u)$ and to the cubic $u^3 - 1.4u^2 + 0.4u$, the second varies with U_1, with g_{21} proportional to $\cos(4\pi u)$, the other components are zero, and the signal to noise ratio is 4. The uneven cell keeps this structure, the cosine and the ratio, and replaces the sine by a Gaussian bump of standard deviation 0.10 at 0.40 minus 0.8 times one of standard deviation 0.018 at 0.72, and the cubic by the chirp $\sin\{2\pi(u + 3.5u^2)\}$, whose frequency goes

from one to eight cycles over the interval; both are multiplied by an infinitely differentiable window that equals one on $[0.08, 0.92]$ and vanishes with all its derivatives at 0 and 1 (Section S9 of the Supplementary Material). These components are infinitely differentiable, within the smoothness conditions of the spline theory of Xue and Yang (2006), but the scale of two of them changes along the interval. The inhomogeneous cell keeps the structure, with the bumps, blocks and heavisine functions of Donoho and Johnstone (1994) in place of the sine, the cubic and the cosine, and signal to noise ratio 3. The mixed cell places the components of the inhomogeneous one in a model with $(p, q) = (4, 4)$, so that 3 of its 16 blocks are active, with the same ratio. In the null cell every component is zero and the model is a linear regression with $(p, q) = (3, 2)$; the standard deviation of the error is fixed at 0.62, about a quarter of that of the regression function.

In the terms of Assumption 5, the effective regularity is $s' = 3/2$ in the smooth cell, where the cubic has different slopes at the two ends of the interval and its periodic extension has a corner, and $s' = 1/2$ in the inhomogeneous and mixed cells, where blocks and heavisine jump inside the interval. The components of the uneven cell have infinitely differentiable periodic extensions, and their regularity is limited only by the vanishing moments of the wavelet.

Each cell is run at $n = 250$, 500, 1000 and 2000, with 100 replicates of each pair of cell and sample size. A replicate consists of a training sample of size n, a test sample of size 2000 from the same law, with errors at the standard deviation of the training sample, and a partition of the training sample into 10 folds; the three are shared by every method, so that the comparisons within a replicate are paired.

Three arms change one factor at a time in the inhomogeneous and the mixed cells, at $n = 1000$ and with 100 replicates each. In the first, the signal to noise ratio is 1 instead of 3. In the second, the modulators are coupled by a Gaussian copula with correlation 0.5 between every pair, which keeps the uniform marginals and makes the cross terms of the Gram matrix between distinct modulators nonzero; the density of this copula is unbounded at two corners of the unit square and vanishes at the other two, so that this arm lies outside Assumption 3. In the third arm the linear covariates depend on the modulating ones: each non-constant covariate is drawn as $X_\ell = (1-\rho^2)^{1/2}\zeta_\ell + \rho\, h(U_{\ell-1})$ with $\rho = 0.5$, where ζ_ℓ is standard normal and $h(u) = \sqrt{12}\,(u - 1/2)$ is the standard score of a uniform modulating covariate, so that X_2 is paired with U_1, the modulating covariate of its own coefficient. Each X_ℓ keeps unit variance and has correlation ρ with its pair, and given $\mathbf{U}$ the non-constant covariates have covariance $(1-\rho^2)I$, so that the eigenvalue

condition of Assumption 2 holds while the Gram matrix of the products no longer factorizes as a Kronecker product. Gaussian covariates are not bounded, in this arm as in every cell, so that the design lies outside the bound B_X of Assumption 2; what the third arm exercises is the eigenvalue condition, the part of that assumption that allows the covariates to depend on the modulators. A replicate of an arm is drawn from the seeds of the same replicate of its core cell, and the first and the third arms share with it the modulators, the draws ζ_ℓ and the standardized errors.

A larger model [E4.3: run if the pilot measures its cost below about ten hours of processor time] has $(p, q) = (6, 4)$ at $n = 1000$, with 24 blocks of which 6 are active: the active structure of the inhomogeneous cell twice, on $(X_1, X_2) \times (U_1, U_2)$ and on $(X_3, X_4) \times (U_3, U_4)$, with β_5 and β_6 constant and signal to noise ratio 3.

Every sample is derived from a single master seed, through an injective encoding of the cell, the sample size, the replicate and the stream. The seeds are independent of those of the pilot runs on which the estimator, its tuning rules and the grids below up to $n = 1000$ were settled, so that the study does not evaluate the method on the samples on which it was chosen.

5.2 Methods and tuning

Every method is tuned in its own terms, by the criteria it is used with, so that a difference between two methods is not a difference of tuning. The WAFC is the estimator (2.5) with $b = \lceil \log n \rceil$ and weights $|G|^{1/2}$, on the periodized extremal phase wavelets of Daubechies (1992) with four vanishing moments ($L = 8$), evaluated by table and with no margin; (J, λ) is chosen by 10-fold cross-validation (Section 4.3) and the threshold by the one standard error rule on the same folds (Section 4.4), the default of the software. The row WAFC (cv) comes from the same search, with the threshold that minimizes the cross-validated error, the option aimed at prediction. Seven other rows enter the tables, five competitors and two references.

GAM (REML) and GAM (GCV) fit the additive varying coefficient model by penalized regression splines with the `mgcv` package (Wood, 2017, 2025), through its function `bam` (Wood et al., 2015), with one smooth of U_m multiplied by X_ℓ for every block and an extra penalty on the null space of each smooth, which lets a smooth be shrunk to zero (Marra and Wood, 2011). The first chooses the smoothing parameters by restricted maximum likelihood and the second by generalized cross-validation, and each chooses the basis dimension k, common to the smooths, from a grid by its own

criterion, which is the full search of Ruppert (2002) and the search of Pya and Wood (2016, Section 2.3). GAM (GCV) is not run in the mixed cell or in the larger model, where a single fit at the top of its grid takes more than an hour.

The B-spline group lasso expands each component in cubic B-splines, centered on the training sample, and fits a group lasso (Yuan and Lin, 2006) with one group per block and the levels unpenalized, with `grpreg`; the number of basis functions per block, 2^J, and the penalty are chosen by cross-validation on the folds. It differs from the WAFC in the basis and in the unit of the penalty, a whole block instead of a chunk.

The adaptive spline gives each block a cubic spline with its own knots, chosen from the data, in the spirit of Wang et al. (2024), who choose the knots by dynamic programming in a model with a single modulator. Here the knots of a block, at most 12 among 40 candidates at quantiles of its modulator, are inserted one at a time by the Bayesian information criterion, inside a backfitting cycle over the blocks, so that this row stands in for their method and does not implement it.

The block lasso (KP) is the estimator of Klopp and Pensky (2015) on the wavelet design of the WAFC, with their chunk size $\lceil \log n \rceil$ and the levels penalized as one more group, as in their norm. It departs from their

estimator in the chunks, which stay inside a resolution level, each coarse level being a chunk of its own, and in the weights, which are the $|G|^{1/2}$ of the solver; it differs from the WAFC in the penalized levels and in the cut of the chunks. Its (J, λ) is chosen by cross-validation, with no threshold.

The two references are the oracle support, the estimator (2.5) on the columns of the active blocks only, with (J, λ) chosen by cross-validation and no threshold, whose distance to the WAFC is the price of not knowing the structure; and the constant coefficients, least squares on $\mathbf{X}$ alone, which is the correct model in the null cell, where the oracle support reduces to it, and a floor in the others. The coordinatewise lasso of Section 4, at the pair chosen by cross-validation and thresholded by the one standard error rule, is reported in Section S9 of the Supplementary Material.

The grids were fixed by the pilot runs and do not depend on the cell: $J \in \{2, \ldots, 8\}$ for the methods on the wavelet design, the same J for the 2^J B-splines per block, and $k \in \{5, 10, 20, 40, 80\}$ for the two searches of `mgcv`, one grid for both criteria, which avoids powers of two so that the splines are not tuned on the grid of the wavelets. A resolution is capped by the number of distinct points of the modulator: at $n = 250$ the 255 wavelets of $J = 8$ exceed the 249 distinct points that a modulator has on the circle after rescaling, where its two extremes meet at $0 \equiv 1$, so the level is built as

$J = 7$ and the effective grid there is $\{2, \dots, 7\}$, and the dimension 256 of the B-splines is dropped. At $n = 2000$, and only there, the grids are extended before the study, to $J \in \{2, \dots, 9\}$ for the methods on the wavelet design and for the B-splines, and with $k = 120$ added to the grid of `mgcv`, because the resolution that the theory prescribes grows with n, as in (3.4).

5.3 Measures

Every method is asked the same questions on the test sample and on a grid of 256 equally spaced points in the observed range of each modulator. The prediction error is the root mean squared difference between the fitted and the true regression function on the test sample. The error of the components is the integrated squared error summed over the blocks, with the fit and the truth both centered on the grid, because every method fixes the level of a component by its own convention; it is split between the active blocks and the null ones, where it measures what a method puts where there is nothing. The two errors are reported separately, since a method can predict well with wrong components, or the reverse. The structure is read from the blocks a method keeps, as the frequency $\mathrm{P}(\widehat{\mathcal{S}} = \mathcal{S})$ of the replicates in which they are exactly the active ones, beside the mean numbers of null blocks kept and of active blocks dropped; a smooth of `mgcv` counts as kept when

its effective degrees of freedom exceed 0.1, since the extra penalty shrinks it toward zero but never to zero exactly. The time is the elapsed time of a fit, the search over the grids and the threshold included, in a single process [E4.4: the processor]. Means over the replicates are reported with their standard errors; since the methods share the samples of a replicate, the Supplementary Material also gives the median, within replicate, of the ratio of each error to that of the WAFC, and the fraction of replicates in which a method has the smaller error.

5.4 Results

Table 1 gives the errors and the recovery of the structure at $n = 1000$, and Figure 1 the fitted components of the inhomogeneous cell. [E4.4: the reading of Table 1 and Figure 1.] Section S9 of the Supplementary Material gives the errors at every sample size, the arms [E4.3: and the larger model], the structure, the resolution and the dimension chosen with the frequency of choices at the top of a grid, the time, and the slope of the logarithm of the errors against $\log n$. [E7: the address of the compendium that reproduces the study.]

Table 1: Simulation at $n = 1000$, 100 replicates per cell. [E4.4: root mean squared prediction error and integrated squared error of the components, mean and standard error, by method and cell, and the frequency with which the blocks kept are exactly the active ones.]

[E4.4: the table, from `results/tables/`.]

6. Application

6.1 Data and model

At a roadside site the total oxidant OX = $NO_2 + O_3$ is close to linear in the concentration of the nitrogen oxides NOx (Clapp and Jenkin, 2001, Section 2.3): the intercept is the regional background of oxidant, and the slope is the local contribution to it, which Carslaw and Beevers (2005, Section 2.4) use as an estimate of the fraction of NOx emitted directly as NO_2, the primary fraction. The slope is an estimate and not the fraction itself: it also carries the NO_2 formed by the thermal reaction of NO with O_2 and from emitted nitrous acid, and it is lower at night (Clapp and Jenkin, 2001, Section 2.3). At Marylebone Road, London, Carslaw (2005, Sections 3.1 and 3.2, Figure 3(a)) estimated the primary fraction at about 10% by volume from 1997 to 2002, rising through 2002 and 2003 to about 23% at the end of 2003, and related the rise mainly, though not wholly, to

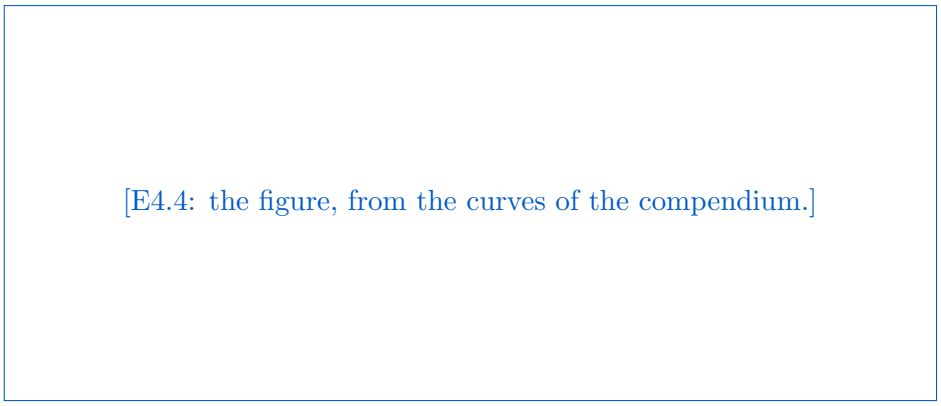


Figure 1: Inhomogeneous cell, $n = 1000$: the components g_{11} (bumps, left) and g_{12} (blocks, right), the truth and the fits of the WAFC and of GAM (REML). [E4.4: the line types, the replicate or the pointwise summary over the 100 replicates that is drawn, and the finding.]

the particle filters fitted to the buses (his Sections 3.3 and 3.4). A coefficient of NOx that varies with the date can show such a change without a model for its timing, and a second modulator asks whether the coefficient also moves with the weather.

The data are the hourly concentrations of NOx, expressed as NO_2, of NO_2 and of O_3 at Marylebone Road, site MY1 of the Automatic Urban and Rural Network, from January 1998 to June 2005 (Department for Environment, Food and Rural Affairs, 2026), converted to parts per billion (ppb) with the conversion factors of the same department (Department for Environment, Food and Rural Affairs, 2005), and the hourly wind speed at

10 m of the ERA5 reanalysis (Hersbach et al., 2020, 2023) at the nearest grid point, retrieved through the Open-Meteo service (Zippenfenig, 2024) and interpolated to the middle of each hour. The 61,280 hours with the four variables observed are used. The concentrations are published under the Open Government Licence and the wind under the Creative Commons Attribution 4.0 licence; the attributions they require are given in the Acknowledgements.

The response is OX in ppb, the linear covariates are $X_1 \equiv 1$ and X_2, NOx in units of 100 ppb, so that β_2 reads as a percentage of NOx, and the modulators are the date and the wind speed, each acting on both coefficients, so that $p = q = 2$ and the model has four blocks. Both modulators are mapped to $[0, 1]$ by their range. The date is not periodic, and the periodized basis joins the end of the series to its start (Section 4.2), so that the fits are read away from the first and the last 60 days. The series also has intervals without measurements, the longest of 43 days in the summer of 1998; over those longer than the spacing of the wavelets of the finest level chosen, about 11 days, the estimate is not determined by the data and is not shown.

6.2 Fitting and evaluation

The errors are correlated over hours and days, and a test sample of hours scattered among the training hours would reward a method for learning the level of each week. The unit of every split is therefore a week. The prediction error is estimated over 20 random partitions that hold out 30% of the weeks, and inside each training sample the weeks are spread over 10 folds, which every method tuned by cross-validation uses; the figure shows the fits to the whole sample, with folds formed in the same way. The partitions are drawn from the master seed of the simulation study and are independent of the exploratory partitions on which this data set was chosen. Each method is tuned in its own terms, as in Section 5.2: the WAFC chooses (J, λ), with $J \in \{2, \ldots, 8\}$, and the threshold by cross-validation on these folds, `mgcv` chooses the smoothing parameters and $k \in \{5, 10, 20, 40, 80\}$ by REML or by GCV, and the constant coefficients have nothing to tune. None of the methods models the dependence of the errors. The error of a method is the root mean squared error on the held-out weeks, averaged over the partitions, with the standard error of Nadeau and Bengio (2003), which corrects for the overlap between the training samples of different partitions.

6.3 The step of 2003

Figure 2 shows each coefficient along each modulator, with the component of the other modulator at its sample mean. The background of oxidant follows the annual cycle, over a range of 23 to 24 ppb in the fits to the whole sample. The slope on NOx is flat until mid 2002 and then rises, by 9.0 percentage points from its level before 2002 to its level from 2004 on in the fit to the whole sample, and by 9.4 to 10.3 points over the 20 partitions; it has completed a quarter of the rise by January 2003 (0.17 to 0.36 over the partitions) and 90% of it by the end of July 2003 in the fit to the whole sample and in 18 of the 20 partitions, and by the end of the gap in the data of late July and early August 2003 in the other two. The two thresholds keep the same blocks along the date and give the same rise, and the spline fitted by REML gives a rise of 10.3 points, 90% of it by the end of that gap. At this site Carslaw (2005, Figure 3(a)) estimated a rise of the primary fraction from about 10% to about 23% by volume over 2002 and 2003; the step of the fit is measured from 2004 on, past the peak described next, and the rise to that peak is 13.5 points (11.7 to 13.8 over the partitions). Two departures from a single step appear in every fit of the WAFC and of the spline fitted by REML, a dip at the beginning of 2002 and a peak in the second half of 2003: the dip also appears in the monthly estimates of

Carslaw (2005, Figure 3(a)), whose series ends in December 2003, before the slope falls back from the peak. Neither is interpreted here.

The wind speed modulates the slope by 0.8 to 1.8 percentage points, 9% to 17% of the step, in the fits to the whole sample that keep that block. The WAFC drops that block, in the fit to the whole sample and in every partition, while the threshold of the smallest cross-validated error keeps it in 19 of the 20 partitions and the splines fitted by REML and by GCV in 18 and 19, so the fit does not support the absence of that modulation, only its size, which is small beside the step. Both modulators act on the background in every fit of the threshold of the smallest cross-validated error and of the two splines, the wind speed over a range of 5.7 to 7.8 ppb in the fits to the whole sample; the WAFC keeps the date and drops the wind speed in the fit to the whole sample, and keeps the wind speed in 12 of the 20 partitions. On the held-out weeks, the WAFC with the threshold of the smallest cross-validated error predicts as well as the better of the two splines: its root mean squared error is 11.52 ppb against 11.47 for the spline fitted by REML, which has the smaller mean error of the two; the ratio to the better spline of each partition is 1.005 on average, and the paired difference, 0.05 ppb with standard error 0.11, is negative in 8 of the 20 partitions. The WAFC is behind by 2.5%, with an error of 11.75 ppb

and a paired difference of 0.28 ppb (standard error 0.22). Section S10 of the Supplementary Material gives the errors of every method, the frequency with which each block is kept over the partitions and the choices of J, λ, the threshold and k, and shows that extending the grids of J and k does not change the comparison.

6.4 A second application

Section S10 of the Supplementary Material applies the method to the hourly PM2.5 at the site Dongsi in Beijing, from March 2013 to February 2017, in the Beijing Multi-Site Air Quality data of the UCI Machine Learning Repository (Chen, 2017; Zhang et al., 2017): the logarithm of PM2.5 is regressed on the wind speed, the temperature and the pressure, with coefficients modulated by the day of the year, a periodic modulator, and the relative humidity, over 34,287 hours. The question there is prediction. The coefficients change with the season, around the nominal dates of the heating season, 15 November to 15 March (Liang et al., 2015, Section 7), and on the held-out weeks the WAFC is ahead of the better of the two splines in all 20 partitions, with a root mean squared error 8% below that of the better spline on average.

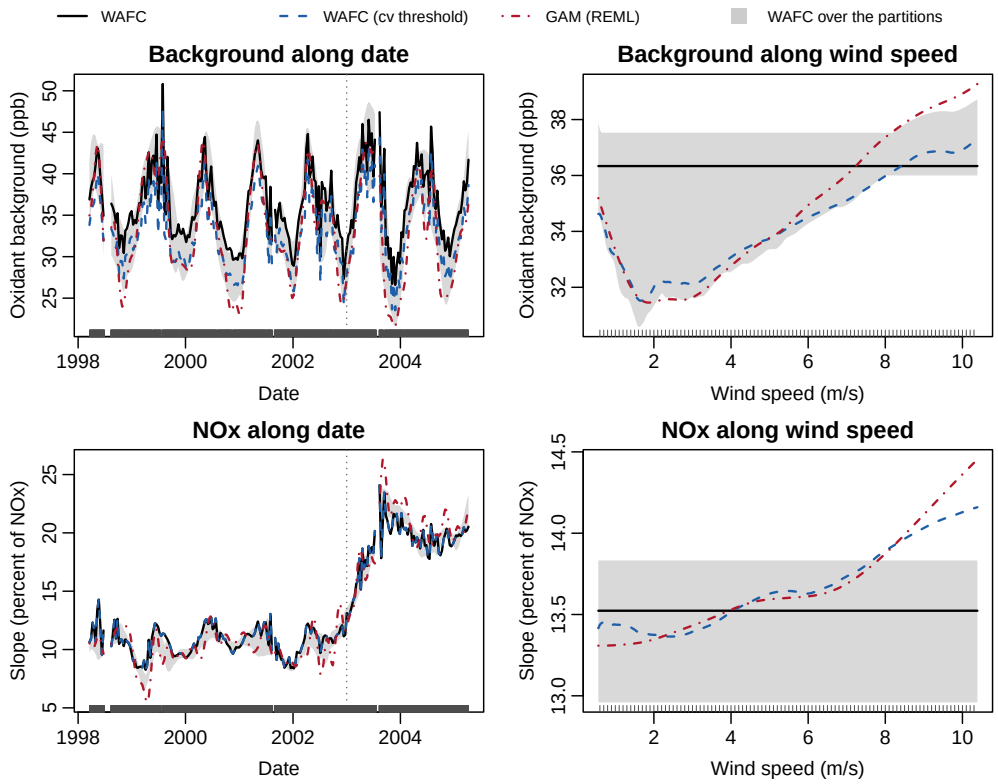


Figure 2: Marylebone Road, London, hourly, 1998–2005: the oxidant background (top, ppb) and the slope on NOx (bottom, percent of NOx) along the date (left) and the wind speed (right), with the component of the other modulator at its sample mean; the band is the range of the WAFC over 20 fits on 70% of the weeks. Not shown: the first and the last 60 days, where the periodized basis joins the end of the series to its start, and the gaps in the data longer than the spacing of the finest wavelets, about 11 days; ticks on the axes mark the observed values.

Supplementary Materials

The online Supplementary Material contains the proofs of all the results stated in the paper, together with the auxiliary lemmas they use: the identification argument and the properties of the periodized basis, the approximation lemmas in L_2 and in the supremum norm and the proposition on the cost of periodization, the population and empirical bounds for the Gram matrix of the product design, the ~~four lemmas~~properties of the partition into chunks, the calibration of the penalty over chunks and the other lemmas behind the oracle inequality and its form for the estimator that the software computes, ~~and~~ the ~~two lemmas~~lemmas behind the rates, ~~and~~ the ~~lemma~~ two lemmas behind the recovery of the block structure by thresholding and the risk of the thresholded estimator, and the theory of the coordinatewise lasso, with its proofs. It also gives the test functions and the larger model of the simulation study, with its results at every sample size and in every arm, and further results of the application, with the second one, to data from Beijing.

Acknowledgements

Write the acknowledgements here.

The data of Section 6 are used under the following attributions. ©

Crown 2026 copyright Defra via uk-air.defra.gov.uk, licenced under the Open Government Licence (OGL). Generated using or contains modified Copernicus Climate Change Service information 2026. Neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains. Hourly ERA5 wind data were retrieved through the Open-Meteo historical weather API (Zippenfenig, 2024; `https://open-meteo.com/`), licensed under CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`); the instantaneous values were linearly interpolated to the middle of each hour. Beijing Multi-Site Air Quality data, UCI Machine Learning Repository (`https://doi.org/10.24432/C5RK5G`), licensed under CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`); only the site Dongsi is used.

References

Amato, U., A. Antoniadis, I. De Feis, and I. Gijbels (2022). Wavelet-based robust estimation and variable selection in nonparametric additive models. *Statistics and Computing 32*(1), 11.

Antoniadis, A. and J. Fan (2001). Regularization of wavelet approximations. *Journal of the American Statistical Association 96*(455), 939–967.

Antoniadis, A., I. Gijbels, and S. Lambert-Lacroix (2014). Penalized estimation in additive

varying coefficient models using grouped regularization. *Statistical Papers 55*(3), 727–750.

Breheny, P. and J. Huang (2015). Group descent algorithms for nonconvex penalized linear and logistic regression models with grouped predictors. *Statistics and Computing 25*(2), 173–187.

Breheny, P. and R. Miller (2026). grpreg: Regularization Paths for Regression Models with Grouped Covariates. R package version 3.6.0, `https://CRAN.R-project.org/package=grpreg`.

Bühlmann, P. and S. van de Geer (2011). *Statistics for High-Dimensional Data: Methods, Theory and Applications*. Berlin: Springer.

Cai, T. T. (1999). Adaptive wavelet estimation: a block thresholding and oracle inequality approach. *The Annals of Statistics 27*(3), 898–924.

Cai, Z., J. Fan, and Q. Yao (2000). Functional-coefficient regression models for nonlinear time series. *Journal of the American Statistical Association 95*(451), 941–956.

Carslaw, D. C. (2005). Evidence of an increasing NO_2/NO_X emissions ratio from road traffic emissions. *Atmospheric Environment 39*(26), 4793–4802.

Carslaw, D. C. and S. D. Beevers (2005). Estimations of road vehicle primary NO_2 exhaust emission fractions using monitoring data in London. *Atmospheric Environment 39*(1), 167–177.

Chen, S. (2017). Beijing Multi-Site Air Quality. UCI Machine Learning Repository, `https:`

//doi.org/10.24432/C5RK5G, accessed 20 September 2026.

Clapp, L. J. and M. E. Jenkin (2001). Analysis of the relationship between ambient levels of O_3, NO_2 and NO as a function of NO_x in the UK. *Atmospheric Environment 35*(36), 6391–6405.

Cohen, A. (2003). *Numerical Analysis of Wavelet Methods*, Volume 32 of *Studies in Mathematics and its Applications*. Amsterdam: Elsevier.

Cohen, A., I. Daubechies, and P. Vial (1993). Wavelets on the interval and fast wavelet transforms. *Applied and Computational Harmonic Analysis 1*(1), 54–81.

Daubechies, I. (1992). *Ten Lectures on Wavelets*, Volume 61 of *CBMS-NSF Regional Conference Series in Applied Mathematics*. Philadelphia: Society for Industrial and Applied Mathematics.

Daubechies, I. and J. C. Lagarias (1991). Two-scale difference equations. I. Existence and global regularity of solutions. *SIAM Journal on Mathematical Analysis 22*(5), 1388–1410.

Department for Environment, Food and Rural Affairs (2005). Conversion factors between ppb and μg m^{-3} and ppm and mg m^{-3}. UK-AIR report 0502160851, https://uk-air.defra.gov.uk/library/reports?report_id=306.

Department for Environment, Food and Rural Affairs (2026). Hourly measurements at Marylebone Road (MY1), Automatic Urban and Rural Network, 1998–2005. UK-AIR, https://uk-air.defra.gov.uk, accessed 5 October 2026, Open Government Licence.

Donoho, D. L. and I. M. Johnstone (1994). Ideal spatial adaptation by wavelet shrinkage. *Biometrika 81*(3), 425–455.

Donoho, D. L. and I. M. Johnstone (1998). Minimax estimation via wavelet shrinkage. *The Annals of Statistics 26*(3), 879–921.

Fan, J. and W. Zhang (2008). Statistical methods with varying coefficient models. *Statistics and Its Interface 1*(1), 179–195.

Friedman, J., T. Hastie, and R. Tibshirani (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software 33*(1), 1–22.

Härdle, W., G. Kerkyacharian, D. Picard, and A. Tsybakov (1998). *Wavelets, Approximation, and Statistical Applications*, Volume 129 of *Lecture Notes in Statistics*. New York: Springer.

Haris, A., N. Simon, and A. Shojaie (2018). Wavelet regression and additive models for irregularly spaced data. In *Advances in Neural Information Processing Systems 31*, pp. 8987–8997. arXiv:1903.04631.

Hastie, T. and R. Tibshirani (1993). Varying-coefficient models. *Journal of the Royal Statistical Society: Series B (Methodological) 55*(4), 757–779.

Hersbach, H., B. Bell, P. Berrisford, G. Biavati, A. Horányi, J. Muñoz Sabater, J. Nicolas, C. Peubey, R. Radu, I. Rozum, D. Schepers, A. Simmons, C. Soci, D. Dee, and J.-N. Thépaut (2023). ERA5 hourly data on single levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), `https://doi.org/10.24381/cds.adbb2d47`, accessed 5 October 2026.

Hersbach, H., B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. De Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R. J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. de Rosnay, I. Rozum, F. Vamborg, S. Villaume, and J.-N. Thépaut (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society 146*(730), 1999–2049.

Hsu, D., S. M. Kakade, and T. Zhang (2012). A tail inequality for quadratic forms of subgaussian random vectors. *Electronic Communications in Probability 17*(52), 1–6.

Huang, J., J. L. Horowitz, and F. Wei (2010). Variable selection in nonparametric additive models. *The Annals of Statistics 38*(4), 2282–2313.

Johnstone, I. M. (2019). Gaussian estimation: Sequence and wavelet models. Draft version of September 16, 2019.

Klopp, O. and M. Pensky (2015). Sparse high-dimensional varying coefficient model: nonasymptotic minimax study. *The Annals of Statistics 43*(3), 1273–1299.

Liang, X., A. Cohen, A. Sólon Heinsfeld, F. Pestilli, and D. J. McDonald (2024). sparsegl: An R package for estimating sparse group lasso. *Journal of Statistical Software 110*(6), 1–23.

Liang, X., T. Zou, B. Guo, S. Li, H. Zhang, S. Zhang, H. Huang, and S. X. Chen (2015). Assessing Beijing's $PM_{2.5}$ pollution: severity, weather impact, APEC and winter heat-

ing. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 471*(2182), 20150257.

Liu, R. and L. Yang (2010). Spline-backfitted kernel smoothing of additive coefficient model. *Econometric Theory 26*(1), 29–59.

Lounici, K., M. Pontil, S. van de Geer, and A. B. Tsybakov (2011). Oracle inequalities and optimal inference under group sparsity. *The Annals of Statistics 39*(4), 2164–2204.

Ma, S., R. J. Carroll, H. Liang, and S. Xu (2015). Estimation and inference in generalized additive coefficient models for nonlinear interactions with high-dimensional covariates. *The Annals of Statistics 43*(5), 2102–2131.

Marra, G. and S. N. Wood (2011). Practical variable selection for generalized additive models. *Computational Statistics & Data Analysis 55*(7), 2372–2387.

Montoril, M. H. (2026). WaveBased: Wavelet-Based Estimation. R package version 2.6-0, `https://github.com/michelcias/WaveBased`.

Montoril, M. H., W. Chang, and B. Vidakovic (2019). Wavelet-based estimation of generalized discriminant functions. *Sankhya B 81*(2), 318–349.

Montoril, M. H., P. A. Morettin, and C. Chiann (2018). Wavelet estimation of functional coefficient regression models. *International Journal of Wavelets, Multiresolution and Information Processing 16*(1), 1850004.

Nadeau, C. and Y. Bengio (2003). Inference for the generalization error. *Machine Learn-*

ing 52(3), 239–281.

Pya, N. and S. N. Wood (2016). A note on basis dimension selection in generalized additive modelling.

R Core Team (2026). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. Version 4.6.1.

Ruppert, D. (2002). Selecting the number of knots for penalized splines. *Journal of Computational and Graphical Statistics 11*(4), 735–757.

Sardy, S. and X. Ma (2024). Sparse additive models in high dimensions with wavelets. *Scandinavian Journal of Statistics 51*(1), 89–108.

Sardy, S. and P. Tseng (2004). AMlet, RAMlet, and GAMlet: automatic nonlinear fitting of additive models, robust and generalized, with wavelets. *Journal of Computational and Graphical Statistics 13*(2), 283–309.

Simon, N., J. Friedman, T. Hastie, and R. Tibshirani (2013). A sparse-group lasso. *Journal of Computational and Graphical Statistics 22*(2), 231–245.

Simon, N. and R. Tibshirani (2012). Standardization and the group lasso penalty. *Statistica Sinica 22*(3), 983–1001.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological) 58*(1), 267–288.

Triebel, H. (1983). *Theory of Function Spaces*, Volume 78 of *Monographs in Mathematics*. Basel:

Birkhäuser. Reimpresso em 2010 na série Modern Birkhäuser Classics, DOI 10.1007/978-3-0346-0416-1.

Wang, X., B. Jiang, and J. S. Liu (2024). Varying coefficient model via adaptive spline fitting. *Journal of Computational and Graphical Statistics 33*(2), 614–624.

Wei, F., J. Huang, and H. Li (2011). Variable selection and estimation in high-dimensional varying-coefficient models. *Statistica Sinica 21*(4), 1515–1540.

Wood, S. (2025). mgcv: Mixed GAM Computation Vehicle with Automatic Smoothness Estimation. R package version 1.9-4, `https://CRAN.R-project.org/package=mgcv`.

Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R* (Second ed.). Boca Raton, FL: Chapman and Hall/CRC.

Wood, S. N., Y. Goude, and S. Shaw (2015). Generalized additive models for large data sets. *Journal of the Royal Statistical Society: Series C (Applied Statistics) 64*(1), 139–155.

Xue, L. and H. Liang (2010). Polynomial spline estimation for a generalized additive coefficient model. *Scandinavian Journal of Statistics 37*(1), 26–46.

Xue, L. and L. Yang (2006). Additive coefficient modeling via polynomial spline. *Statistica Sinica 16*(4), 1423–1446.

Yuan, M. and Y. Lin (2006). Model selection and estimation in regression with grouped variables. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 68*(1), 49–67.

Zhang, S., B. Guo, A. Dong, J. He, Z. Xu, and S. X. Chen (2017). Cautionary tales on air-quality improvement in Beijing. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 473*(2205), 20170457.

Zhang, S. and M.-Y. Wong (2003). Wavelet threshold estimation for additive regression models. *The Annals of Statistics 31*(1), 152–173.

Zhou, X. and J. You (2004). Wavelet estimation in varying-coefficient partially linear regression models. *Statistics & Probability Letters 68*(1), 91–104.

Zippenfenig, P. (2024). Open-Meteo.com Weather API. Zenodo, version 1.4.0, `https://doi.org/10.5281/zenodo.7970649`.

first author affiliation

E-mail: (first author email)

second author affiliation

E-mail: (second author email)